

Project Title: Student Success & Career Guidance Portal

Platform: Salesforce Education Cloud

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Problem Statement

Colleges and universities struggle with tracking **student performance, internships, and placements**. Data is scattered across multiple systems (LMS, placement cells, manual records), making it difficult to create a **360° student profile**.

Students receive **generic career advice** instead of personalized guidance. Institutions lack **data-driven insights** to improve curriculum and placement strategies.

Proposed Salesforce Solution

Building a **Salesforce Education Cloud app** that:

- Centralizes student academic, extracurricular, and internship data.
 - Integrates with LMS to track real-time progress.
 - Leverages **Einstein AI** to recommend personalized career paths, internships, and certifications.
 - Provides dashboards for faculty, placement officers, and management to measure student success and employability.
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Project Overview

The *Student Success & Career Guidance Portal* is a Salesforce Education Cloud CRM application designed to help colleges and universities track **student performance, skills, internships, and placements** in one centralized system. The CRM provides a **360° view of each student**, integrates learning management features, and supports placement officers with real-time analytics and dashboards. Using **Einstein AI**, it can recommend career paths and internships that match student skills, improving overall employability.

Key features include:

- Centralized **student records** with academic details and skills.
 - **Internship & Placement tracking** with company mapping.
 - **Job Sync Flow** to fetch job posts automatically into Salesforce.
 - Reports & dashboards for placement officers to monitor student readiness.
 - Role-based security to ensure data confidentiality and controlled access.
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Objectives

The primary goal of this CRM is to **bridge the gap between students, faculty, and placement officers** by providing a single platform for student success tracking.

Specific objectives include:

- **Student Success Monitoring:** Track GPA, skills, and readiness scores for career guidance.
- **Internship & Placement Management:** Streamline the process of managing job offers and company interactions.
- **Automation & Integration:** Use Salesforce Flows to automate repetitive tasks such as job synchronization, skill updates, and career recommendations.
- **Analytics for Decision-Making:** Provide dashboards showing top skills, placement funnel, and average career readiness for better institutional planning.
- **Scalability & Future Growth:** Ensure the CRM can be extended with new modules (e.g., alumni management, employer feedback) to support long-term needs.

By achieving these objectives, the CRM provides **business value** such as:

- Higher employability for students.
- Streamlined placement operations.

- Data-driven decision-making for colleges.
 - Increased transparency and efficiency across departments.
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Phase 1: Problem Understanding and Industry Analysis

1.1. Problem Statement

Colleges face challenges in:

Tracking student academic performance, co-curricular activities, and skills in one place.

Managing internships and placement drives effectively.

Connecting students with career opportunities based on their skills and interests.

Offering personalized career guidance that improves student employability.

This leads to:

Manual data management.

Missed internship and placement opportunities.

Low student engagement and unclear career paths.

Our project addresses these issues by building a Salesforce Education Cloud App that:

Maintains a complete student profile.

Integrates with LMS.

Tracks skills, certifications, and attendance.

Uses Einstein AI to recommend career paths, internships, and placements.

1.2. Requirement Gathering

Functional Requirements

Student Management: Create student records with personal info, academic history, skills, and certifications.

Company/Internship Management: Track recruiter companies, internship openings, eligibility criteria, and placement status.

Performance Tracking: Map grades, attendance, and event participation.

Career Guidance: Use AI to suggest suitable career paths based on skill gaps and performance.

Notifications: Send alerts for upcoming interviews, deadlines, or recommended internships.

Non-Functional Requirements

Scalability: Must handle thousands of students and company records.

Data Security: Follow Salesforce's security model (profiles, roles, sharing rules).

User Experience: Offer a simple Lightning UI for students, faculty, and placement officers.

Integration: Ensure smooth integration with external LMS and job portals using REST APIs.

1.3. Stakeholder Analysis

Stakeholder	Role	Expectations
College Admin/Placement Officer	Manages student data, tracks placements	Easy dashboard, scheduling tools, analytics
Faculty	Inputs grades, monitors performance	Simple UI to update student progress
Students	View performance, apply for internships	Personalized career guidance, job alerts
Recruiter Companies	Post internships, hire students	Access to eligible candidates, communication
Management	Decision-making based on reports	Complete view of performance and placement stats

1.4. Business Process Mapping

As-Is Process (Current Situation):

Student data maintained manually (Excel and spreadsheets).

Placement drives coordinated via email and phone.

No centralized performance dashboard.

Students depend on generic career advice.

To-Be Process (After Salesforce Implementation):

Student data stored in Salesforce (centralized).

Automatic flow for internships (eligibility check, scheduling, and status tracking).

Real-time dashboards for college management.

1.5. Industry-Specific Use Case Analysis

Education Industry Trends:

Growing use of Education CRM platforms for student lifecycle management.

Need for data-driven career guidance to enhance placement rates.

Demand for personalized learning paths using AI and machine learning.

Salesforce is widely used in education for:

Student information systems.

Engagement and retention.

Career counseling and alumni management.

Our solution fits these industry needs perfectly.

1.6. AppExchange Exploration

In this phase, we explored Salesforce AppExchange apps such as:

TargetX and EDA (Education Data Architecture): For student lifecycle management.

Blackthorn Events: For scheduling placement drives.

Mulesoft Connectors: For integrating external LMS.

We will use Salesforce Education Data Architecture (EDA) as the base model and customize it for our project.

1.7. Use Cases, Student Success & Career Guidance Portal

Student Management

- Create and maintain student records, including personal info, academic history, skills, and certifications.
- Centralize all performance data in one place for faculty, placement officers, and students to access.
- Track attendance, grades, and participation in events or workshops.

Company & Internship Management

- Maintain a database of recruiter companies with details such as industry, open roles, and locations.
- Record internship and job postings with eligibility criteria, stipend, and duration.
- Track student applications and connect them to companies.

Placement & Internship Scheduling

- Schedule interviews and placement drives for eligible students.
- Send SMS and email notifications and reminders to students about upcoming events.
- Track student attendance and results from each placement round.

Career Guidance & Recommendations

- Use Einstein AI to evaluate student performance and skills.
- Recommend career paths, certifications, and internships based on skill gaps.
- Display recommended opportunities on the student portal or dashboard.

Reporting & Analytics

- Provide a dashboard for placement officers that shows:
 - Number of students placed
 - Internship success rates
 - Skills gap analysis across batches
 - Track student progress over semesters.
 - Offer management real-time insights for decision-making.
-

Phase 2 — Org setup & configuration

2.1. Basic org settings

1. **Setup** → *Company Settings* → **Company Information** → update Default Locale, Time Zone, Default Currency.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'company' and a list of settings categories: Company Settings (expanded), Calendar Settings, Company Information (selected), Data Protection and Privacy, Fiscal Year, Holidays, Language Settings, and My Domain. The main content area is titled 'Company Information' and includes a 'Zip/Postal Code' field. Below this are three sections: 'Locale Settings' with dropdowns for Default Locale (English (India)), Default Language (English), and Default Time Zone ((GMT+05:30) India Standard Time (Asia/Kolkata)); 'Currency Settings' with a dropdown for Currency Locale (English (United States) - USD); and 'Translation Settings' with a checkbox for Enable Data Translation. A yellow warning banner states: 'Turning on multiple currencies introduces permanent changes in your org. This feature can't be turned off. Review the Implications of Enabling Multiple Currencies before enabling.' Below this is a checkbox for 'Activate Multiple Currencies'. At the bottom, there is a 'Salesforce Newsletter Settings' section.

2. **Setup** → *Business Hours* → **New** → create Placement Cell Hours.

Q company

Company Settings

- Business Hours
- Calendar Settings
 - Public Calendars and Resources
- Company Information
- Data Protection and Privacy
- Fiscal Year
- Holidays
- Language Settings
- My Domain

Didn't find what you're looking for? Try using Global Search.

Business Hours

Step 1. Business Hours Name

Business Hours Name Use these business hours as the default ☐

Active ☐

Step 2. Time Zone

Time Zone

Step 3. Business Hours

Sunday	9:00 AM	to	5:00 PM	☐ 24 hours
Monday	9:00 AM	to	5:00 PM	☐ 24 hours
Tuesday	9:00 AM	to	5:00 PM	☐ 24 hours
Wednesday	9:00 AM	to	5:00 PM	☐ 24 hours
Thursday	9:00 AM	to	5:00 PM	☐ 24 hours
Friday	9:00 AM	to	5:00 PM	☐ 24 hours
Saturday	9:00 AM	to	5:00 PM	☐ 24 hours

Save Cancel

3. **Setup** → *Holidays* → add semester breaks.

Q holiday

Company Settings

- Holidays

Didn't find what you're looking for? Try using Global Search.

Holidays

Holidays are dates and times at which business hours are suspended. Business hours are the days and hours that your support team is available.

Holidays [New](#)

Action	Holiday Name	Description	Date and Time
Edit Del	Diwali Holiday	There is an diwali holiday	10/20/2025 All Day
Edit Del	Ganesh Visharjan Holiday	There is an holiday for Ganesh Visharjan	9/19/2025 All Day

Elapsed Holidays

Action	Holiday Name	Description	Date and Time
Clone	Semester Break	This is a Semester Break in our college	9/11/2025 All Day

2.2 Enable features

1. **Setup** → *Quick Find* → **Digital Experiences** → enable if you want a student/recruiter portal.

2.3 Users, Profiles, Roles

1. **Setup** → *Users* → **New User** → create sample accounts: student1@, faculty1@, placement1@.

SETUP
Users

To get more licenses, use the Your Account app. [Let's Go](#)

View: All Users [Edit](#) [Create New View](#)

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**

[New User](#) [Reset Password\(s\)](#) [Add Multiple Users](#)

☐ Action	Full Name ↑	Alias	Username	Role	Active	Profile
☐ Edit	Chatter Expert	Chatter	chatty.00dgl000007qlcfuag.fgnblrmn2d0c@chatter.salesforce.com		☒	Chatter Free User
☐ Edit Login	EPIC_OrgFarm	OEPIQ	epic.eb76301f20c2@orgfarm.salesforce.com		☒	Test Profile
☐ Edit Login	Kostha, Prashant	pkost	faculty1@demo.com		☒	Faculty Profile
☐ Edit	Nema, Aris	anema	student1@demo.com		☐	Student Profile
☐ Edit Login	Sethi	seth	placement1@demo.com		☒	Placement Profile
☐ Edit	Tamrakar, Swayam	cse	cse22_swayamtamrakar887@agentforce.com		☒	System Administrator
☐ Edit	User_Integration	integ	integration@00dgl000007qlcfuag.com		☒	Analytics Cloud Integration User
☐ Edit	User_Security	sec	insightssecurity@00dgl000007qlcfuag.com		☒	Analytics Cloud Security User

[New User](#) [Reset Password\(s\)](#) [Add Multiple Users](#)

A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | Other | **All**

2. **Setup** → *Profiles* → open Standard User → **Clone** → name Student_Profile; repeat for Faculty_Profile, Placement_Profile. Set baseline object permissions (Contacts: read for Students, edit for Faculty).
3. **Setup** → *Roles* → **Set Up Roles** → create hierarchy:
 - Admin
 - Placement Head
 - Placement Officer
 - Faculty
 - Faculty Member

- Student



4. **Setup** → *Permission Sets* → **New** → Placement_Officer_PS — grant object and field-level permissions not in profile (e.g., export data).

SETUP Permission Sets

Permission Set **Placement Officer** Video Tutorial | Help for this Page

Find Settings... [Clone](#) [Edit Properties](#) [Manage Assignments](#) [View Summary](#)

Permission Set Overview

Description	Extra permissions for placement officers (data export, placements).	API Name	Placement_Officer
License		Namespace Prefix	
Session Activation Required	☐	Created By	Swayam Tamrakar, 9/12/2025, 12:38 AM
Permission Set Groups Added To	0	Last Modified By	Swayam Tamrakar, 9/12/2025, 12:54 AM

Apps

- Assigned Apps**
Settings that specify which apps are visible in the app menu
- Assigned Connected Apps**
Settings that specify which connected apps are visible in the app menu
- Object Settings**
Permissions to access objects and fields, and settings such as tab availability
- App Permissions**
Permissions to perform app-specific actions, such as "Manage Call Centers"

2.4 Security / OWD / Sharing

1. **Setup** → *Sharing Settings* → set **Contact** OWD = **Private** (students' personal data protected).

The screenshot shows the 'Sharing Settings' page in Salesforce Setup. At the top, there's a 'SETUP' header with a shield icon and the title 'Sharing Settings'. Below this, a paragraph explains that the page displays organization-wide sharing settings. A dropdown menu 'Manage sharing settings for:' is set to 'All Objects'. A button 'Disable External Sharing Model' is visible. The main section is 'Default Sharing Settings' with a sub-section 'Organization-Wide Defaults' and an 'Edit' button. A table lists default settings for various objects:

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchie:
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Public Read/Write	Private	✓
Contact	Private	Private	✓
Order	Controlled by Parent	Controlled by Parent	✓

2. Create **Sharing Rules**: e.g., Share Student (Contact) records to Role = Placement Officer where Department = Computer Science.

The screenshot shows the 'Sharing Rules' page in Salesforce Setup. On the left, a sidebar contains a search bar with 'shar' and a list of links: 'Security', 'Guest User Sharing Rule Access Report', and 'Sharing Settings' (highlighted). The main area is titled 'Sharing Rules' and lists five categories: 'Lead Sharing Rules', 'Account Sharing Rules', 'Contact Sharing Rules', 'Opportunity Sharing Rules', and 'Case Sharing Rules'. Each category has 'New' and 'Recalculate' buttons and a 'Help' link. The 'Contact Sharing Rules' section is expanded, showing a table with the following details:

Action	Criteria	Shared With	Contact
Edit Del	Contact: Department EQUALS Computer Science	Role: Placement Officer	Contact Read/Write

3. **Setup** → *Login Access Policies* → enable admin access if needed for troubleshooting.

The screenshot shows the Salesforce Setup interface for 'Login Access Policies'. On the left, a navigation menu includes 'Identity' (with 'Login Flows' and 'Login History') and 'Security' (with 'Login Access Policies' highlighted). The main content area is titled 'Login Access Policies' and includes a subtitle 'Control which support organizations your users can grant login access to.' Below this is a 'Manage Support Options' section with 'Save' and 'Cancel' buttons. A table lists settings for 'Administrators Can Log in as Any User' (checked) and 'Support Organization' (Salesforce.com Support). The table has columns for 'Support Organization', 'Packages', 'Available to Users' (radio button selected), and 'Available to Administrators Only' (radio button unselected). A 'Help for this Page' link is in the top right.

Setting	Enabled
Administrators Can Log in as Any User	☒

Support Organization	Packages	Available to Users	Available to Administrators Only
Salesforce.com Support		☒	☐

2.5 Dev org & sandbox usage

- Use Developer org for building, Sandbox for UAT. Create a Partial Copy sandbox for more realistic data if available.
 - Dev Org Setup
 - To implement the following project a Salesforce Developer Edition org was setup.
 - Made a GitHub Repository for source control.
 - Setting up the VS Code and SFDX for the implementation of the LWC Component for development.
-

Phase 3 — Data modeling & relationships

3.1 Use standard objects vs custom

- **Account** → Use for College and Recruiter Company.
- **Contact** → Use for Student (recommended).
- Create custom objects:
 - Skill__c (label: Skill)

The screenshot shows the Salesforce Setup > Object Manager interface for a custom object named "Skill". The left sidebar contains a navigation menu with the following items: Details (selected), Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Triggers, Flow Triggers, and Validation Rules.

The main content area is titled "Skill" and contains the following sections:

- Custom Object Information**: This section includes instructions on using singular and plural labels. It contains input fields for "Label" (Skill) and "Plural Label" (Skills), both marked as required. An example "Account" is shown for the label and "Accounts" for the plural label. There is also a checkbox for "starts with vowel sound" which is currently unchecked.
- The Object Name is used when referencing the object via the API.**: This section contains an input field for "Object Name" (Skill) and an example "Account".
- Description**: A large text area for the object's description.
- Context-Sensitive Help Setting**: Two radio buttons are present: "Open the standard Salesforce.com Help & Training window" (selected) and "Open a window using a Visualforce page".
- Content Name**: A dropdown menu currently set to "None".
- Enter Record Name Label and Format**: This section includes instructions on the Record Name. It contains an input field for "Record Name" (Skill Name) and an example "Account Name".
- Data Type**: A dropdown menu set to "Text". A warning message states: "Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type."
- Optional Features**: A list of checkboxes: "Allow Reports" (checked), "Allow Activities" (checked), and "Track Field History" (checked).

- Student_Skill__c (junction between Contact & Skill)

SETUP > OBJECT MANAGER

Student Skill

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports. Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label Example: Account

Plural Label Example: Accounts

starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Data Type Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Display Format Example: A-{0000} What Is This?

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History

- Internship__c

SETUP > OBJECT MANAGER

Internship

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports. Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label Example: Account

Plural Label Example: Accounts

starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Data Type Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History
- ☐ Allow in Chatter Groups

- Placement__c

SETUP > OBJECT MANAGER

Placement

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules

Custom Object Information ⓘ Required Information

The singular and plural labels are used in tabs, page layouts, and reports.
Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label Example: Account

Plural Label Example: Accounts

starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Data Type Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History

- Career_Recommendation__c

SETUP > OBJECT MANAGER

Career Recommendation

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules

The singular and plural labels are used in tabs, page layouts, and reports.
Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label Example: Account

Plural Label Example: Accounts

starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Data Type Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Display Format Example: A-0000 [What Is This?](#)

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History

- (Optional) Job_Post__c for external job postings

The screenshot shows the Salesforce Setup > Object Manager interface for the 'Job Post' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Triggers, Flow Triggers, and Validation Rules. The main content area is titled 'Custom Object information' and includes the following sections:

- Custom Object information:** A note states, 'The singular and plural labels are used in tabs, page layouts, and reports. Be careful when changing the name or label as it may affect existing integrations and merge templates.' It contains input fields for 'Label' (Job Post, Example: Account) and 'Plural Label' (Job Posts, Example: Accounts). A checkbox for 'Starts with vowel sound' is present.
- The Object Name is used when referencing the object via the API:** An input field for 'Object Name' (Job_Post, Example: Account) and a large text area for 'Description'.
- Context-Sensitive Help Setting:** Radio buttons for 'Open the standard Salesforce.com Help & Training window' (selected) and 'Open a window using a Visualforce page'.
- Content Name:** A dropdown menu set to 'None'.
- Enter Record Name Label and Format:** A note explains that the Record Name appears in page layouts, key lists, related lists, lookups, and search results. It includes an input field for 'Record Name' (Job Post Name, Example: Account Name) and a 'Data Type' dropdown set to 'Text'. A warning states: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.'
- Optional Features:** A list of checkboxes: 'Allow Reports' (checked), 'Allow Activities' (checked), 'Track Field History' (checked), and 'Allow in Chatter Groups' (unchecked).

3.2 Create custom objects and fields

For each object:

1. **Setup** → *Object Manager* → **Create** → **Custom Object** → fill:
 - Label: Internship → Plural: Internships → Object Name: Internship__c → Check “Allow Reports”, “Allow Activities”, “Track Field History” as needed → Save.
2. **Fields & Relationships** → **New**:
 - Example Internship__c fields:
 - Company__c — Lookup(Account) → Select *Data Type*: Lookup(Account).

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field Information

Field Label	Company	Object Name	Internship
Field Name	Company	Data Type	Lookup
API Name	Company__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Swayam Tamrakar, 9/13/2025, 10:33 AM	Modified By	Swayam Tamrakar, 9/13/2025, 10:33 AM

Lookup Options

Related To	Account	Child Relationship Name	Internships
Related List Label	Internships		
Required	☐		
What to do if the lookup record is deleted?	Clear the value of this field.		

Lookup Filter

No lookup filters defined.

Validation Rules

New

[Validation Rules Help](#) ?

No validation rules defined.

- Student__c — Lookup(Contact).

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field Information

Field Label	Student	Object Name	Internship
Field Name	Student	Data Type	Lookup
API Name	Student__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Swayam Tamrakar, 9/13/2025, 10:34 AM	Modified By	Swayam Tamrakar, 9/13/2025, 10:34 AM

Lookup Options

Related To	Contact	Child Relationship Name	Internships
Related List Label	Internships		
Required	☐		
What to do if the lookup record is deleted?	Clear the value of this field.		

Lookup Filter

No lookup filters defined.

Validation Rules

New

[Validation Rules Help](#) ?

No validation rules defined.

▪ Start_Date__c — Date

Internship Custom Field

Start Date

[Back to Internship](#)

[Validation Rules \(0\)](#)

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field Information

Field Label	Start Date	Object Name	Internship
Field Name	Start_Date	Data Type	Date
API Name	Start_Date__c		
Description	Internship Starting date		
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Swayam Tamrakar, 9/11/2025, 4:19 AM	Modified By	Swayam Tamrakar, 9/11/2025, 4:19 AM

General Options

Required ☒

Default Value

Validation Rules

New

[Validation Rules Help](#)

No validation rules defined.

[Back To Top](#)

Always show me [more](#) records per related list

▪ End_Date__c — Date

Internship Custom Field

End Date

[Back to Internship](#)

[Validation Rules \(0\)](#)

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field Information

Field Label	End Date	Object Name	Internship
Field Name	End_Date	Data Type	Date
API Name	End_Date__c		
Description	Internship Ending Date		
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Swayam Tamrakar, 9/11/2025, 4:19 AM	Modified By	Swayam Tamrakar, 9/11/2025, 4:19 AM

General Options

Required ☒

Default Value

Validation Rules

New

[Validation Rules Help](#)

No validation rules defined.

[Back To Top](#)

Always show me [more](#) records per related list

- Status__c — Picklist values: Applied, Interviewing, Selected, Completed, Rejected.

Internship Custom Field
Status
[Back to Internship](#)

[Validation Rules](#) [0]

Custom Field Definition Detail [Edit](#) [Set Field-Level Security](#) [View Field Accessibility](#) [Where is this used?](#)

Field Information

Field Label	Status	Object Name	Internship
Field Name	Status	Data Type	Picklist
API Name	Status__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Swayam Tamrakar, 9/13/2025, 10:42 AM	Modified By	Swayam Tamrakar, 9/13/2025, 10:42 AM

General Options

Required	☒
Default Value	i

Picklist Options

Restrict picklist to the values defined in the value set	☒
Controlling Field	[New]

Picklist Values Used

Active and inactive picklist values	5 (1,000 max)
-------------------------------------	---------------

- Stipend__c — Currency

Superior
[Back to Internship](#)

[Validation Rules](#) [0]

Custom Field Definition Detail [Edit](#) [Set Field-Level Security](#) [View Field Accessibility](#) [Where is this used?](#)

Field Information

Field Label	Stipend	Object Name	Internship
Field Name	Stipend	Data Type	Currency
API Name	Stipend__c		
Description			
Help Text			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			
Created By	Swayam Tamrakar, 9/13/2025, 10:43 AM	Modified By	Swayam Tamrakar, 9/13/2025, 10:43 AM

General Options

Required	☒
Default Value	

Currency Options

Length	18
Decimal Places	0

Validation Rules [New](#) [Validation Rules Help](#)

No validation rules defined.

[^ Back To Top](#) Always show me [more](#) records per related list

- **Role__c** — *Text* or *Picklist* (role title)

SETUP > OBJECT MANAGER
Internship

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Flow Triggers

Validation Rules

Field Label: **Role**

Values: ☐ Use global picklist value set ☒ Enter values, with each value separated by a new line

Software Engineer
Civil Engineer
Mechanical Engineer
Electrical Engineer
Computer Engineer
Chemical Engineer
Industrial Engineer

☒ Display values alphabetically, not in the order entered

☐ Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field Name: **Role**

Description:

Help Text:

Required: ☒ Always require a value in this field in order to save a record

Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity

Default Value: [Show Formula Editor](#)

Einstein

3. Create Skill__c:

- Field Name used for skill label (e.g., “Java”).

Skill Field

Skill Name

[Back to Skill](#)

[Set Field-Level Security](#) [View Field Accessibility](#)

Field Information

Field Label	Skill Name	Field Name	Name
Data Type	Text(80)		
Description			
Data Owner			
Field Usage			
Data Sensitivity Level			
Compliance Categorization			

Validation Rules [New](#) [Validation Rules Help](#)

No validation rules defined.

- Skill_Level__c — Picklist: Beginner, Intermediate, Advanced.

Step 2. Enter the details Step 2 of 4

Previous Next Cancel

Field Label ⓘ

Values

☐ Use global picklist value set

☒ Enter values, with each value separated by a new line

☐ Display values alphabetically, not in the order entered

☐ Use first value as default value ⓘ

☒ Restrict picklist to the values defined in the value set ⓘ

Field Name ⓘ

Description

Help Text

Required ☒ Always require a value in this field in order to save a record

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity ⓘ

4. Create junction Student_Skill__c:

- Student__c — Lookup(Contact)

SETUP > OBJECT MANAGER

Student Skill

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Flow Triggers

Validation Rules

NEW Relationship

Step 3. Enter the label and name for the lookup field Step 3 of 6

Previous Next Cancel

Field Label ⓘ

Field Name ⓘ

Description

Help Text

Child Relationship Name ⓘ

Required ☐ Always require a value in this field in order to save a record

What to do if the lookup record is deleted?

☒ Clear the value of this field. You can't choose this option if you make this field required.

☐ Don't allow deletion of the lookup record that's part of a lookup relationship.

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity ⓘ

Lookup Filter

Optionally, create a filter to limit the records available to users in the lookup field. [Tell me more!](#)

▶ [Show Filter Settings](#)

- Skill__c — Lookup(Skill)

SETUP > OBJECT MANAGER
Student Skill

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Validation Rules

Student Skill
New Relationship

Step 3. Enter the label and name for the lookup field Step 3 of 6

Field Label Skill
Field Name Skill
Description
Help Text

Child Relationship Name Student_Skills
Required ☒ Always require a value in this field in order to save a record
What to do if the lookup record is deleted? ☐ Clear the value of this field. You can't choose this option if you make this field required.
☒ Don't allow deletion of the lookup record that's part of a lookup relationship.
Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Lookup Filter
Optionally, create a filter to limit the records available to users in the lookup field. [Tell me more!](#)
[Show Filter Settings](#)

- Proficiency__c — Number or Picklist as needed

SETUP > OBJECT MANAGER
Student Skill

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Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
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Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Validation Rules

Student Skill
New Relationship

Step 2. Enter the details Step 2 of 4

Field Label Proficiency
Values ☐ Use global picklist value set
☒ Enter values, with each value separated by a new line
Beginner
Intermediate
Advanced
☐ Display values alphabetically, not in the order entered
☐ Use first value as default value
☒ Restrict picklist to the values defined in the value set
Field Name Proficiency
Description
Help Text

Required ☒ Always require a value in this field in order to save a record
Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

5. Placement__c

- Student__c — *Lookup(Contact)*

The screenshot shows the 'New Relationship' configuration page for the 'Student__c' field. The left sidebar lists various configuration options, with 'Fields & Relationships' selected. The main content area is titled 'Step 3. Enter the label and name for the lookup field'. The 'Field Label' and 'Field Name' are both set to 'Student'. The 'Child Relationship Name' is 'Placements'. The 'Required' checkbox is checked. The 'What to do if the lookup record is deleted?' options are 'Don't allow deletion of the lookup record that's part of a lookup relationship.' (selected) and 'Clear the value of this field. You can't choose this option if you make this field required.' The 'Auto add to custom report type' checkbox is also checked. A 'Lookup Filter' section is present at the bottom with a link to 'Show Filter Settings'.

SETUP > OBJECT MANAGER
Placement

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Validation Rules

Placement
New Relationship

Step 3. Enter the label and name for the lookup field

Field Label: Student
Field Name: Student
Description:
Help Text:
Child Relationship Name: Placements
Required: ☒ Always require a value in this field in order to save a record
What to do if the lookup record is deleted?:
☒ Don't allow deletion of the lookup record that's part of a lookup relationship.
☐ Clear the value of this field. You can't choose this option if you make this field required.
Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity
Lookup Filter
Optionally, create a filter to limit the records available to users in the lookup field. [Tell me more!](#)
[Show Filter Settings](#)

- Company__c — *Lookup(Account)*

The screenshot shows the 'New Relationship' configuration page for the 'Company__c' field. The left sidebar lists various configuration options, with 'Fields & Relationships' selected. The main content area is titled 'Step 3. Enter the label and name for the lookup field'. The 'Field Label' and 'Field Name' are both set to 'Company'. The 'Child Relationship Name' is 'Placements'. The 'Required' checkbox is checked. The 'What to do if the lookup record is deleted?' options are 'Don't allow deletion of the lookup record that's part of a lookup relationship.' (selected) and 'Clear the value of this field. You can't choose this option if you make this field required.' The 'Auto add to custom report type' checkbox is also checked. A 'Lookup Filter' section is present at the bottom with a link to 'Show Filter Settings'.

SETUP > OBJECT MANAGER
Placement

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules
Scoping Rules
Object Access
Triggers
Flow Triggers
Validation Rules

Placement
New Relationship

Step 3. Enter the label and name for the lookup field

Field Label: Company
Field Name: Company
Description:
Help Text:
Child Relationship Name: Placements
Required: ☒ Always require a value in this field in order to save a record
What to do if the lookup record is deleted?:
☒ Don't allow deletion of the lookup record that's part of a lookup relationship.
☐ Clear the value of this field. You can't choose this option if you make this field required.
Auto add to custom report type: ☒ Add this field to existing custom report types that contain this entity
Lookup Filter
Optionally, create a filter to limit the records available to users in the lookup field. [Tell me more!](#)
[Show Filter Settings](#)

○ Offer_Package__c — Currency

SETUP > OBJECT MANAGER

Placement

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Flow Triggers

Validation Rules

Step 2. Enter the details

Step 2 of 4

Previous Next Cancel

Field Label

Offer Package

Please enter the length of the number and the number of decimal places. For example, a number with a length of 8 and 2 decimal places can accept values up to "12345678.90".

Length

18

Decimal Places

0

Number of digits to the left of the decimal point

Number of digits to the right of the decimal point

Field Name

Offer_Package

Description

Enter your offer package

Help Text

Required

☐ Always require a value in this field in order to save a record

Auto add to custom report type

☒ Add this field to existing custom report types that contain this entity

Default Value

Show Formula Editor

Use formula syntax: Enclose text and picklist value API names in double quotes: ("The text"). Include numbers without quotes: (25). Show percentages as decimals: (0.10), and express date calculations in the standard format: (Today() + 7). To reference a field from a Custom Metadata type record use: \$CustomMetadata.Type__mdt.RecordAPIName.Field__c

Einstein

○ Offer_Status__c — Picklist: Offered, Accepted, Declined, Joined, Withdrawn

Custom Field Definition Detail

Field Information

Field Label	Offer Status	Object Name	Offer
Field Name	Offer_Status	Data Type	Picklist
API Name	Offer_Status__c		
Description			
Help Text			
Data Owner			
Field Length			
Data Sensitivity Level			
Compliance Categorization			
Created By	System Administrator	Modified By	System Administrator
	9/11/2025, 9:18 AM		9/11/2025, 9:18 AM

General Options

Required ☐

Default Value

Picklist Options

Exempt picklist values from the value set ☒

Controlling Field

Picklist Values Used

Active and Inactive picklist values 4 (1,000 max)

Field Dependencies

No dependencies defined.

Validation Rules

No validation rules defined.

Values

Action	Value	API Name	Default	Chart Column	Modified By
☐ Edit ☐ Delete	Offered	Offered	☐	Assigned dynamically	System Administrator 9/11/2025, 9:18 AM
☐ Edit ☐ Delete	Accepted	Accepted	☐	Assigned dynamically	System Administrator 9/11/2025, 9:18 AM
☐ Edit ☐ Delete	Rejected	Rejected	☐	Assigned dynamically	System Administrator 9/11/2025, 9:18 AM
☐ Edit ☐ Delete	Placed	Placed	☐	Assigned dynamically	System Administrator 9/11/2025, 9:18 AM

- **Joining_Date__c** — *Date*

SETUP > OBJECT MANAGER

Placement

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Sharing Rules

Step 2. Enter the details Step 2 of 4

Previous Next Cancel

Field Label

Field Name

Description

Help Text

Required ☒ Always require a value in this field in order to save a record

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Default Value

Use formula syntax: Enclose text and picklist value API names in double quotes: ("the_text"), include numbers without quotes: (25), show picklist values: (picklist_value), reference a field: (field_name)

Einstein

- **Recruiter__c** — *Lookup(User)* (optional)

SETUP > OBJECT MANAGER

Placement

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Step 3. Enter the label and name for the lookup field Step 3 of 5

Previous Next Cancel

Field Label

Field Name

Description

Help Text

Child Relationship Name

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Lookup Filter

6. E. Career_Recommendation__c

- **Student__c** — *Lookup(Contact)*

The screenshot shows the Salesforce Setup page for the **Career Recommendation** object. The left sidebar contains a navigation menu with the following items: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Restriction Rules. The main content area is titled "Step 3. Enter the label and name for the lookup field" and "Step 3 of 6". It includes a "Previous" button, a "Next" button, and a "Cancel" button. The form fields are: Field Label (Student), Field Name (Student), Description (empty), and Help Text (empty). Below these fields is a section for "Child Relationship Name" with the value "Career_Recommendations". There are three checkboxes: "Required" (checked), "What to do if the lookup record is deleted?" (radio button selected for "Don't allow deletion of the lookup record that's part of a lookup relationship."), and "Auto add to custom report type" (checked).

- **Recommended_Path__c** — *Text* (e.g., “Data Science – Junior Analyst”)
- **Confidence_Score__c** — *Number* (0 - 100)
- **Generated_On__c** — *DateTime*
- **Source__c** — *Picklist*: Einstein, RuleEngine, Manual

7. F. Job_Post__c (optional)

- **Company__c** — *Lookup(Account)*

- **Title__c** — Text

SETUP > OBJECT MANAGER

Job Post

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Restriction Rules

Previous Next Cancel

Field Label ⓘ

Please enter the maximum length for a text field below.

Length

Field Name ⓘ

Description

Help Text ⓘ

Required ☐ Always require a value in this field in order to save a record

Unique ☐ Do not allow duplicate values

☒ Treat "ABC" and "abc" as duplicate values (case insensitive)
☐ Treat "ABC" and "abc" as different values (case sensitive)

External ID ☐ Set this field as the unique record identifier from an external system

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity ⓘ

- **Description__c** — Long Text Area
- **Location__c** — Text
- **Skills_Required__c** — Long Text or multi-select picklist
- **Posted_On__c** — Date

3.3 Record Types & Page Layouts

1. **Contact Object** → *Record Types* → **New** → Student. Add page layouts.

SETUP > OBJECT MANAGER
Contact

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Hierarchy Columns

Step 1. Enter the details Step 1 of 2

Enter a name and description for the new record type. The new record type will include all the picklist values from the existing record type selected below. After saving the new record type, you will be able to customize the picklist values.

Record Type Required Information

Existing Record Type: **--Master--**
Record Type Label: **Student**
Record Type Name: **Student**
Description:
Active: ☒

Select Make Available to give users assigned to this profile the ability to create and clone records of this record type, or assign this record type to existing records. To make the new record type the default for a profile, select Make Default. Users assigned to this record type can still view and edit records associated with record types not available for their profiles.

Profile Name	Record Types Currently Available	☒ Make Available	☐ Make Default
Analytics Cloud Integration User		☒	☐
Analytics Cloud Security User		☒	☐
Anypoint Integration		☒	☐

2. **Page Layouts** → include related lists: Internships, Student_Skills, Placements, Career_Recommendations. Drag fields like GPA__c, Career_Readiness__c.

SETUP > OBJECT MANAGER
Contact

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Hierarchy Columns

Save Quick Save Preview As... Cancel Undo Redo Layout Properties

Fields Quick Find Field Name

Section	Asst. Phone	Contact Owner	Data.com Key	Email	Fax Opt Out	Important	Last Modified By	Level
Blank Space	Birthdate	Contact Record Type	Department	Email Opt Out	Gender Identity	Important Contact	Last Stay-in-Touch...	Mails
Account Name	Buyer Attributes	Created By	Description	Enrollment Year	GPA	Individual	Last Stay-in-Touch...	Major
Assistant	Clean Status	Creation Source	Do Not Call	Fax	Home Phone	Languages	Lead Source	Mobi

Custom Buttons

Fields (Header not visible)
Name Sarah Sample
Account Name Sample Text

Custom Links (Header visible on detail only)

Academic Info
GPA 11.89
Enrollment Year 1,500
Major Sample Text

Mobile Cards (Salesforce mobile only)

SETUP > OBJECT MANAGER
Contact

Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Hierarchy Columns
Scoping Rules
Object Access
Triggers

Save Quick Save Preview As... Cancel Undo Redo Layout Properties

Related Lists Quick Find Related List Name

Activity History	Career Recommenda...	Enrollments	Internships	Open Activities	Related Content	Work Orders
Approval History	Cases	Files	Messaging Sessions	Opportunities	Service Contracts	
Assets	Contact Deliverables	Groups	Messaging Users	Pending Order Bu...	Student Skills	
Campaign History	Data Integration ...	HTML Email Status	Notes & Attachments	Placements	Voice Calls	

Placements Placement Name Sample Text

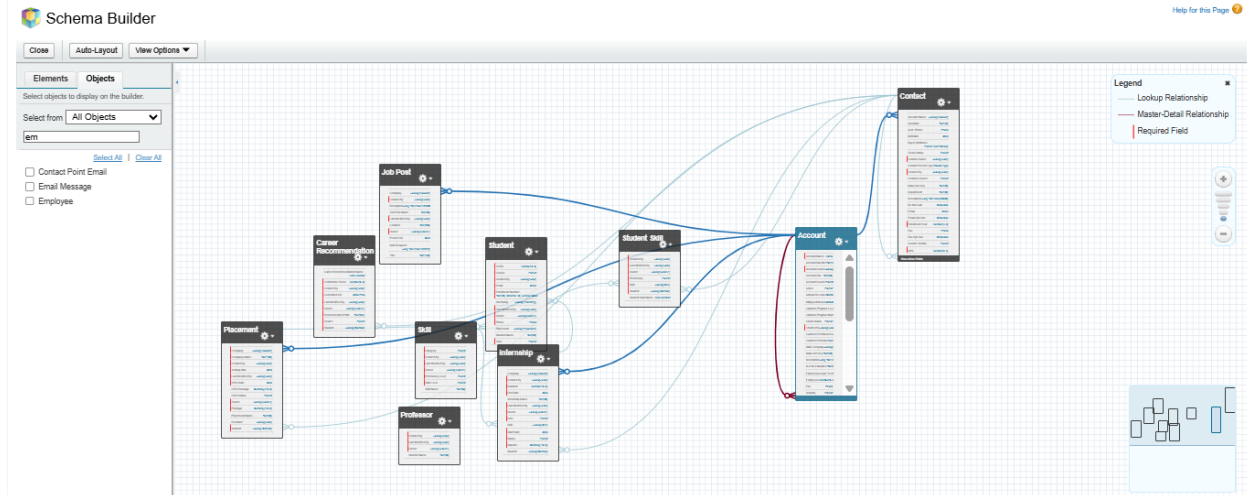
Career Recommendations Career Recommendation Name Sample Text

Student Skills Student Skill Name Sample Text

Internships Internship Name Sample Text

3.4 Schema Builder

1. **Setup** → *Schema Builder* → drag objects to visualize relationships. Confirm junction object links.



3.5 Create sample data to test (validate your model)

- App Launcher → Accounts → New → create XYZ Engineering College.
 - App Launcher → Contacts → New → create Swayam Tamrakar, select Record Type = Student, link Account = XYZ Engineering College, set GPA__c value.
 - App Launcher → Skills tab → New → create Java, Python.
 - App Launcher → Student Skills → New → select Student = Swayam, Skill = Java, set Proficiency.
 - App Launcher → Internships → New → Student = Swayam, Company = TechCorp Pvt Ltd.
 - Open Swayam's Contact page — confirm Student Skills, Internships appear in Related Lists.
-

Phase 4: Process automation

- Use **Flow Builder** for modern automations

4.1 Validation Rules (examples)

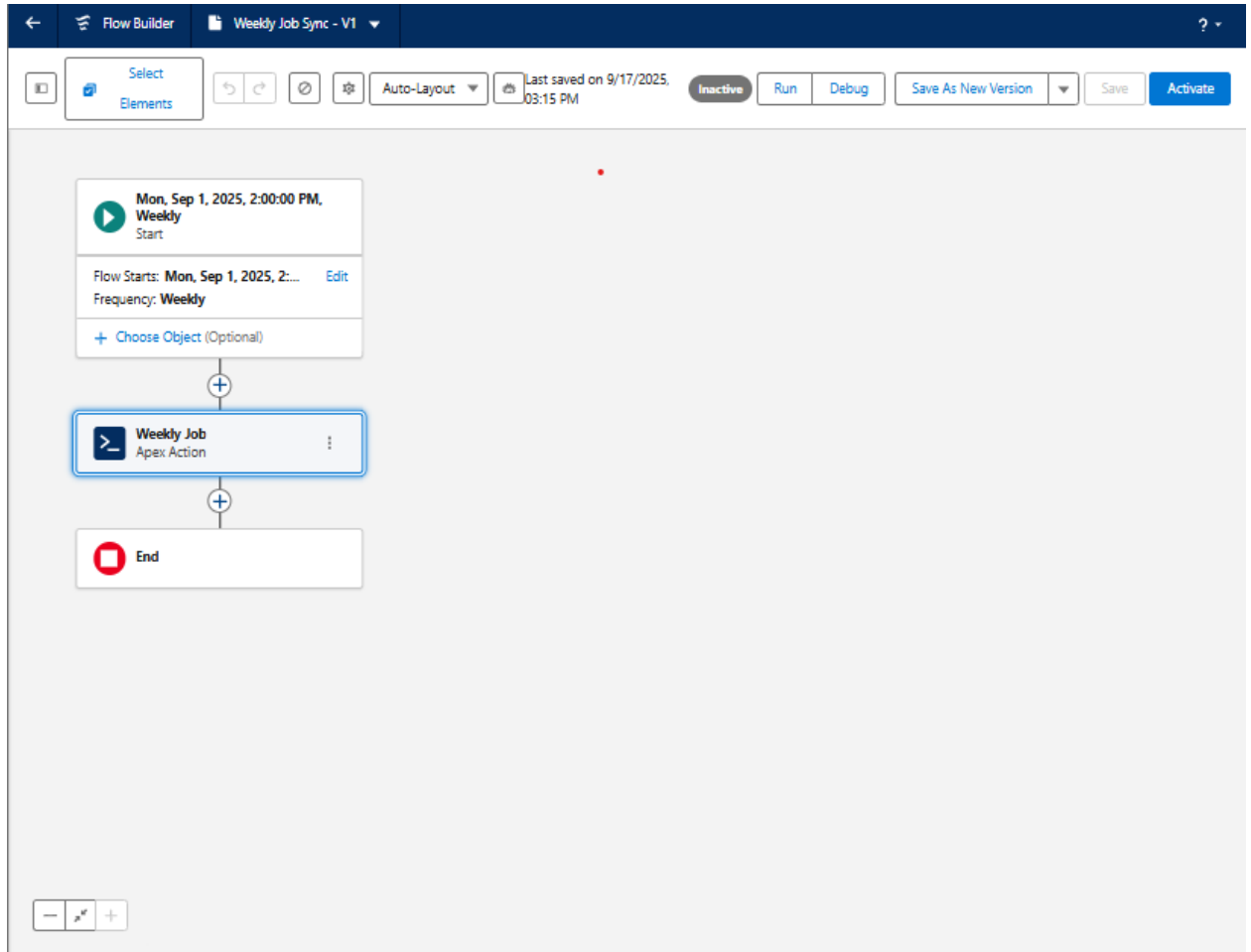
1. **Contact** → *Validation Rules* → **New**

- Rule: GPA__c >= 0 && GPA__c <= 10
- Formula: OR(GPA__c < 0, GPA__c > 10)
- Error message: “Enter GPA between 0 and 10.”

The screenshot shows the Salesforce Validation Rule Editor for the Contact object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Hierarchy Columns, Scoping Rules, Object Access, Triggers, Flow Triggers, and Validation Rules (which is currently selected). The main area is titled 'Validation Rule Edit' and includes a 'Rule Name' field set to 'GPA Check', an 'Active' checkbox, and a 'Description' field. Below this is the 'Error Condition Formula' section, which includes an 'Example' (Discount Percent <= 30%), a 'More Examples...' link, and a note that the formula must be true for the error message to appear. The formula field contains 'OR (GPA__c < 0, GPA__c > 10)'. To the right of the formula field is a 'Functions' panel with a search bar and a list of functions including ABS, ACOS, ADDMONTHS, AND, ASCII, ASIN, and others. Below the formula field is a 'Check Syntax' button. The 'Error Message' section includes an 'Example' (Discount percent cannot exceed 30%), a note that the message appears when the formula is true, and a text field containing 'Enter GPA between 0 and 10'. At the bottom, the 'Error Location' is set to 'Field' with 'GPA' selected. The interface includes 'Save', 'Save & New', and 'Cancel' buttons at the top and bottom right.

4.2 Scheduled Flow — Weekly job aggregator

1. **New** → **Schedule-Triggered Flow** → run weekly.
2. Action: call an **Apex Action** (Batch job) or External Service to fetch jobs, create Job_Post__c records.



4.3 Approval Process (resume approval)

1. **Setup** → *Process Automation* → **Approval Processes** → New → **Jump Start Wizard**
2. Object: Contact.
3. **Name:**
 - Process Name: Student Resume Approval
 - Unique Name: Student_Resume_Approval
4. **Entry Criteria:**
 - Field: Resume_Submitted__c (Checkbox)

- Operator: Equals
 - Value: TRUE
5. **Approver:** Select **Automatically assign to user** → Pick your **Placement Officer** user.
 6. **Email Template:** You can skip or create a quick text-based template under Setup → Classic Email Templates.
 7. **Initial Submission Action:** Field Update → set Resume_Status__c = Submitted
 8. **Final Approval Action:** Field Update → Resume_Status__c = Approved.
 9. **Final Rejection Action:** Field Update → Resume_Status__c = Rejected.
 10. **Save** → **Activate**

The screenshot shows the 'Approval Processes' setup page in Salesforce. The process name is 'Student Resume Approval' and it is active. The unique name is 'Student_Resume_Approval'. The description is 'Contact: Resume Submitted SIGNALS True'. The entry criteria is 'Contact: Resume Submitted SIGNALS True'. The record editability is 'Administrator ONLY'. The approval assignment email template is 'Contact Owner'. The initial submission actions include 'Record Lock' and 'Field Update' (Resume_Submitted). The approval steps include 'Step 1' with 'Assigned Approver' 'User:Seeyam Tammaraj'. The final approval actions include 'Record Lock' and 'Field Update' (Resume_Approval). The final rejection actions include 'Record Lock' (Unlock the record for editing).

4.4 Email Templates & Notifications

❖ Lightning templates.

1. **In your org, click App Launcher → Search "Email Templates" → New Email Template**
2. **Fill details:**
 - **Name:** Placement_Ready_Notification.
 - **Related Entity Type:** Contact (since student is a Contact).
 - **Subject:** Student {{Recipient.Name}} is Placement Ready
 - **Folder:** Keep in Public Email Templates or create a folder Placement Notifications.
 - **In Email Content (HTML):**
Hello {{Recipient.FirstName}},

Student {{Contact.Name}} from {{Contact.Account.Name}} has been marked as Placement Ready.

Career Readiness Score: {{Contact.Career_Readiness__c}}

Please review their profile and recommend to recruiters.

Regards,

Student Success Portal

3. Save

The screenshot shows the Salesforce Email Template Builder interface. At the top, there's a navigation bar with a search bar and various icons. Below it, a breadcrumb trail shows 'Student Success' > 'Home' > 'Home' > 'Reports' > 'Dashboards' > 'Contacts' > 'Students' > 'Internships' > 'Placement_Ready_Notification'. The main header of the template editor shows 'Email Template Placement_Ready_Notification' with 'Edit', 'Clone', and 'Delete' buttons. The 'Details' tab is active, showing 'Information' and 'Message Content' sections. The 'Information' section includes fields for 'Email Template Name' (Placement_Ready_Notification), 'Description', 'Made in Email Template Builder' (checkbox), 'Related Entity Type' (Contact), 'Folder' (Public Email Templates), and 'Enhanced Letterhead'. The 'Message Content' section shows the subject line 'Student {{Recipient.Name}} is Placement Ready' and the body content: 'Hello {{Recipient.FirstName}},', 'Student {{Contact.Name}} from {{Contact.Account.Name}} has been marked as Placement Ready.', 'Career Readiness Score: {{Contact.Career_Readiness__c}}', 'Please review their profile and recommend to recruiters.', 'Regards,', and 'Student Success Portal'. The 'Additional Information' section at the bottom shows 'Created By' (Swayam Tamrakar, 9/17/2025, 11:01 AM) and 'Last Modified By' (Swayam Tamrakar, 9/17/2025, 11:09 AM). At the bottom of the interface, there are links for 'Recent Items' and 'History'.

4.5 Workflow Rules, Field Updates & Tasks

1. Setup → Quick Find → Workflow Rules → New Rule.
2. Choose Object = Contact.
3. Rule Name = High GPA Alert.
4. Evaluation Criteria = "created, and every time it's edited to meet criteria".
5. Rule Criteria → Example: GPA__c >= 9.

6. Save

The screenshot shows the 'Edit Rule High GPAAlert' form within a 'Workflow Rules' setup page. The form is divided into several sections: 'Edit Rule', 'Evaluation Criteria', and 'Rule Criteria'. In the 'Edit Rule' section, the 'Object' is set to 'Contact', the 'Rule Name' is 'High GPAAlert', and the 'Description' field is empty. The 'Evaluation Criteria' section has three radio button options for when to evaluate the rule, with the third option 'created, and any time it's edited to subsequently meet criteria' selected. The 'Rule Criteria' section shows a dropdown set to 'criteria are met' and a table with five rows for defining criteria. The first row is populated with 'Contact: GPA', 'greater than', and '9'. The other four rows are empty with '--None--' in the first two columns. The table is connected by 'AND' logic. At the bottom, there is an 'Add Filter Logic...' link and 'Save' and 'Cancel' buttons.

Workflow Rules

Edit Rule High GPAAlert

Enter the name, description, and criteria to trigger your workflow rule. In the next step, associate workflow actions with this workflow rule.

[Save](#) [Cancel](#)

Edit Rule Help for this Page ?

Object: Contact

Rule Name: High GPAAlert

Description:

Evaluation Criteria

Evaluate the rule when a record is:

- ☐ created
- ☐ created, and every time it's edited
- ☒ created, and any time it's edited to subsequently meet criteria

How do I choose?

Rule Criteria

Run this rule if the criteria are met :

Field	Operator	Value	
Contact: GPA	greater than	9	AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND
--None--	--None--		AND

[Add Filter Logic...](#)

[Save](#) [Cancel](#)

7. Add Workflow Action → choose:

- New Email Alert (send an email to Placement Officer).
- New Field Update (set Placement_Ready__c = TRUE).
- New Task (assign follow-up task to Placement Officer).

8. Activate the workflow rule.

The screenshot shows the Salesforce Setup page for Workflow Rules. The page title is 'Workflow Rules' with a 'SETUP' icon. Below the title, there's a 'Workflow Rule' section for 'High GPA Alert'. A yellow banner at the top right says 'Go with the flow! With Flow Builder, the future of low-code automation, you can do everything you do with workflow rules — and more! Salesforce plans to retire workflow rules and recommends building automation in Flow Builder. [Tell Me More](#) | [Migrate your workflow rules to flows](#)'. Below this, the 'Workflow Rule Detail' section shows the rule is 'Active' (checked), 'Object' is 'Contact', and 'Evaluation Criteria' is 'Evaluate the rule when a record is created, and any time it's edited to subsequently meet criteria'. The 'Rule Criteria' is 'Contact: GPA GREATER THAN 9'. The 'Created By' is 'Swayam Tamrakar, 9/18/2025, 12:42 PM' and 'Modified By' is 'Swayam Tamrakar, 9/18/2025, 12:57 PM'. Below this, the 'Workflow Actions' section shows 'Immediate Workflow Actions' with a table of actions: 'Task' (Review Student for Placement Readiness), 'Email Alert' (Email alert to placement officer), and 'Field Update' (Field Update). There are also 'Time-Dependent Workflow Actions' with a link to 'See an example'. A yellow warning box at the bottom says 'You cannot add new time triggers to an active rule. [Deactivate This Rule](#)'. There are 'Edit', 'Clone', and 'Deactivate' buttons at the top of the rule detail section.

Workflow Rule Detail [Edit](#) [Clone](#) [Deactivate](#)

Rule Name	High GPA Alert	Object	Contact
Active	✓	Evaluation Criteria	Evaluate the rule when a record is created, and any time it's edited to subsequently meet criteria
Description			
Rule Criteria	Contact: GPA GREATER THAN 9		
Created By	Swayam Tamrakar, 9/18/2025, 12:42 PM		Modified By Swayam Tamrakar, 9/18/2025, 12:57 PM

Workflow Actions [Edit](#)

Immediate Workflow Actions

Type	Description
Task	Review Student for Placement Readiness
Email Alert	Email alert to placement officer
Field Update	Field Update

Time-Dependent Workflow Actions [See an example](#)

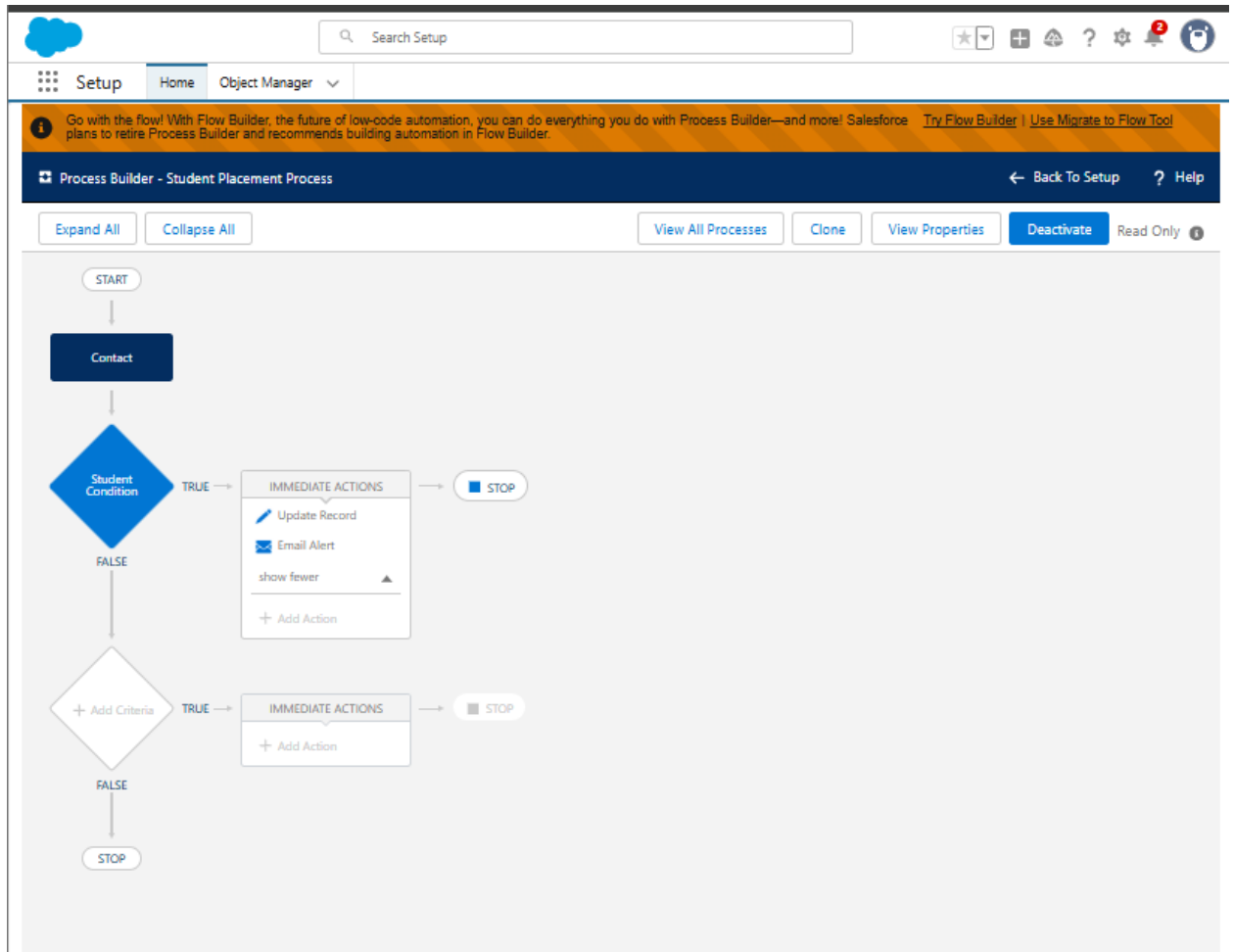
You cannot add new time triggers to an active rule. [Deactivate This Rule](#)

[Edit](#)

4.6 Process Builder

1. **Setup** → Quick Find → **Process Builder** → New.
2. Name: Student Placement Process.
3. Object: **Contact**. Start the process "when a record changes".
4. Criteria: GPA__c >= 8 AND Placement_Status__c = 'Not Ready'.
5. Actions:
 - **Immediate Action** → Update Records → Set Placement_Status__c = Ready.
 - **Immediate Action** → Send Alert → High GPA Alert

6. Activate process.



Phase 5: Apex development

5.1 Trigger + Service + Test

1. Create the Service Class (StudentCareerService)

- **In Developer Console** → **File** → **New** → **Apex Class**.
- **Enter name:** StudentCareerService.
- **Code:**

```
public with sharing class StudentCareerService {
    public static void updateReadiness(Map<Id, Contact> newContactsMap) {
        if (newContactsMap == null || newContactsMap.isEmpty()) return;
        Set<Id> studentIds = new Set<Id>();
        for(Contact c : newContactsMap.values()){
            // adjust 'Student' to your actual RecordType DeveloperName
            if (c.RecordTypeId != null &&
                Schema.SObjectType.Contact.getRecordTypeInfosById()
                    .get(c.RecordTypeId)
                    .getDeveloperName() == 'Student') {
                studentIds.add(c.Id);
            }
        }
        if(studentIds.isEmpty()) return;
        // Aggregate skills
        Map<Id, Integer> skillCounts = new Map<Id, Integer>();
        for (AggregateResult ar : [
            SELECT Student__c studentId, COUNT(Id) cnt
            FROM Student__Skill__c
            WHERE Student__c IN :studentIds
            GROUP BY Student__c
        ]) {
            skillCounts.put((Id) ar.get('studentId'), (Integer) ar.get('cnt'));
        }
        // Calculate readiness
        List<Contact> updates = new List<Contact>();
        for(Contact s : [SELECT Id, GPA__c, Career_Readiness__c
            FROM Contact
            WHERE Id IN :studentIds]){
            Integer skills = skillCounts.containsKey(s.Id) ? skillCounts.get(s.Id) : 0;
            Decimal gpa = (s.GPA__c == null) ? 0 : s.GPA__c;
            Decimal readiness = Math.min(100, (gpa * 8) + (skills * 4));
```

```

        // Only update if value has actually changed
        if(s.Career_Readiness__c != readiness){
            updates.add(new Contact(Id = s.Id, Career_Readiness__c = readiness));
        }
    }
    if(!updates.isEmpty()) update updates;
}
}

```

The screenshot shows the Salesforce Apex Class Detail page for the class **StudentCareerService**. The page includes a header with a "SETUP" button and a "Help for this Page" link. Below the class name, there are tabs for "Apex Class Detail", "Edit", "Delete", "Download", "Security", and "Show Dependencies". The class is currently "Active".

Name	Status
StudentCareerService	Active

Namespace Prefix	Code Coverage
	0% (0/24)

Created By	Last Modified By
Swayam Tamrakar , 9/19/2025, 2:03 AM	Swayam Tamrakar , 9/20/2025, 10:23 AM

Below the metadata, there are tabs for "Class Body", "Class Summary", "Version Settings", and "Trace Flags". The "Class Body" tab is selected, showing the following Apex code:

```

1 public with sharing class StudentCareerService {
2
3     public static void updateReadiness(Map<Id, Contact> newContactsMap) {
4         if (newContactsMap == null || newContactsMap.isEmpty()) return;
5
6         Set<Id> studentIds = new Set<Id>();
7         for(Contact c : newContactsMap.values()){
8             // adjust 'Student' to your actual RecordType DeveloperName
9             if (c.RecordTypeId != null &&
10                Schema.SObjectType.Contact.getRecordTypeInfoById()
11                .get(c.RecordTypeId)
12                .getDeveloperName() == 'Student') {
13                 studentIds.add(c.Id);
14             }
15         }
16     }
17 }

```

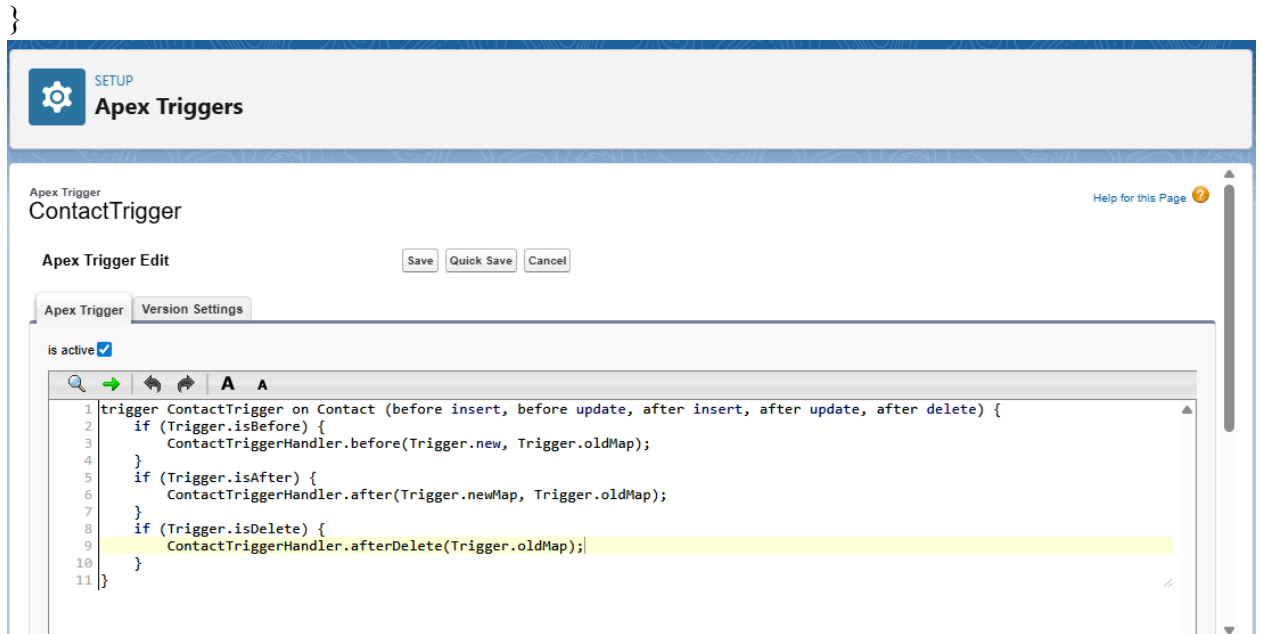
2. Create the Trigger (ContactTrigger)

- **Developer Console** → **File** → **New** → **Apex Trigger**.
- **Object:** Contact. **Name:** ContactTrigger.
- **Code:**

```

trigger ContactTrigger on Contact (before insert, before update, after insert, after update,
after delete) {
    if (Trigger.isBefore) {
        ContactTriggerHandler.before(Trigger.new, Trigger.oldMap);
    }
    if (Trigger.isAfter) {
        ContactTriggerHandler.after(Trigger.newMap, Trigger.oldMap);
    }
    if (Trigger.isDelete) {
        ContactTriggerHandler.afterDelete(Trigger.oldMap);
    }
}

```



3. Create the Test Class

- Developer Console → **File** → **New** → **Apex Class**.

- **Name:** StudentCareerServiceTest.

- **Code:**

@isTest

private class StudentCareerServiceTest {

// ♦ Test Setup - reusable data

@testSetup

static void setup(){

// Create College (Account) for Student

Account college = new Account(Name = 'Test College');

insert college;

// Create Company (Account) for Placement

Account company = new Account(Name = 'TechCorp Pvt Ltd');

insert company;

// Get Student record type

Id studentRtId = Schema.SObjectType.Contact.getRecordTypeInfoByName()
.get('Student').getRecordTypeId();

// Dynamically get a valid Placement_Status__c picklist value (Contact)

Schema.DescribeFieldResult placementStatusField = Contact.Placement_Status__c.
getDescribe();

String validPlacementStatus = placementStatusField.getPicklistValues()[0].getValue

```

0);
    // Create Student
    Contact stud = new Contact(
        LastName = 'TestStudent',
        FirstName = 'Unit',
        AccountId = college.Id,
        GPA__c = 7.5,
        RecordTypeId = studentRtId,
        Enrollment_Year__c = 2025,
        Placement_Status__c = validPlacementStatus
    );
    insert stud;
    // Dynamically get valid Category__c and Proficiency_Level__c for Skill
    Schema.DescribeFieldResult categoryField = Skill__c.Category__c.getDescribe();
    String validCategory = categoryField.getPicklistValues()[0].getValue();

    Schema.DescribeFieldResult proficiencyLevelField = Skill__c.Proficiency_Level__
c.getDescribe();
    String validProficiencyLevel = proficiencyLevelField.getPicklistValues()[0].getVal
ue();
    // Create Skill
    Skill__c skill = new Skill__c(
        Name = 'Apex Programming',
        Category__c = validCategory,
        Proficiency_Level__c = validProficiencyLevel,
        Skill_Level__c = 'Beginner'
    );
    insert skill;
    // Dynamically get a valid Proficiency__c value for Student_Skill__c
    Schema.DescribeFieldResult proficiencyField = Student_Skill__c.Proficiency__c.g
etDescribe();
    String validProficiency = proficiencyField.getPicklistValues()[0].getValue();
    // Link Student with Skill
    Student_Skill__c ss = new Student_Skill__c(
        Student__c = stud.Id,
        Skill__c = skill.Id,
        Proficiency__c = validProficiency
    );
    insert ss;

```

```

        // Dynamically get a valid Offer_Status__c value for Placement
        Schema.DescribeFieldResult offerStatusField = Placement__c.Offer_Status__c.get
Describe();
        String validOfferStatus = offerStatusField.getPicklistValues()[0].getValue();
        // Create Placement linked to Student and Company
        Placement__c placement = new Placement__c(
            Name = 'Software Engineer Placement',
            Company__c = company.Id,          // ✅ Required field
            Company_Name__c = company.Name,
            Joining_date__c = Date.today().addMonths(1),
            Offer_Date__c = Date.today(),
            Offer_Package__c = 600000,
            Offer_Status__c = validOfferStatus,
            Package__c = 600000,
            Student__c = stud.Id
        );
        insert placement;
    }

    // ♦ Test Career Readiness Calculation (Trigger + Service)
    @isTest
    static void testCareerReadinessCalculation(){
        Contact stu = [SELECT Id, GPA__c FROM Contact WHERE LastName = 'TestStu
dent' LIMIT 1];
        // Fire trigger
        stu.GPA__c = 9;
        update stu;
        Contact updatedStu = [SELECT Career_Readiness__c FROM Contact WHERE Id
= :stu.Id];
        System.assert(updatedStu.Career_Readiness__c > 0,
            'Career readiness should be calculated.');
```

```

    }

    // ♦ Test API Callout with Mock (Job Posting Service)
    @isTest
    static void testCalloutWithMockAndFutureMethod(){
        Test.setMock(HttpCalloutMock.class, new MockJobApiResponse());
        Test.startTest();
        JobApiService.fetchJobsAsync(); // assumes you have JobApiService class implem
ented
        Test.stopTest();
    }

```

```

Integer countJobs = [SELECT COUNT() FROM Job_Post__c];
System.assert(countJobs >= 0,
    'Job posts should be inserted from mock response');
}
// ♦ Mock API Response for Job Service
private class MockJobApiResponse implements HttpCalloutMock {
    public HTTPResponse respond(HTTPRequest req){
        HTTPResponse res = new HTTPResponse();
        res.setHeader('Content-Type','application/json');
        res.setBody(['{"id":"123","title":"Intern Software Engineer","company":"TechCo
rp"}']);
        res.setStatusCode(200);
        return res;
    }
}
}

```

4. Run the Test

- In Developer Console → **Test** → **New Run**.
- Select StudentCareerServiceTest.
- Run it.
- Expected Result: Test passes, Coverage shows green for StudentCareerService and ContactTrigger.

5. Verify It Works with Real Data

- In App Launcher → Contacts → New.
- Record Type: Student
- Enter GPA = 7.
- Create a Skill (App Launcher → Skills → New).
- Create Student_Skill (App Launcher → Student Skills → New → link Student + Skill).

- Go back to Student record → refresh → you should see Career Readiness field updated.

5.2 Add Supporting Apex Classes

- Repeat **Setup** → **Apex Classes** → **New** for each:

1. Batch Apex

- **Code:**

global class JobPostingBatch implements Database.Batchable<sObject>, Database.Stateful {

global Database.QueryLocator start(Database.BatchableContext bc){

// Fetch all Accounts

return Database.getQueryLocator([SELECT Id, Name FROM Account]);

}

global void execute(Database.BatchableContext bc, List<sObject> scope){

List<Account> accounts = (List<Account>) scope;

System.debug('Processing Accounts: ' + accounts);

List<Job_Post__c> jobs = new List<Job_Post__c>();

for(Account a : accounts){

jobs.add(new Job_Post__c(

Name = 'Job for ' + a.Name,

Company__c = a.Id, // relates Job Post to Account

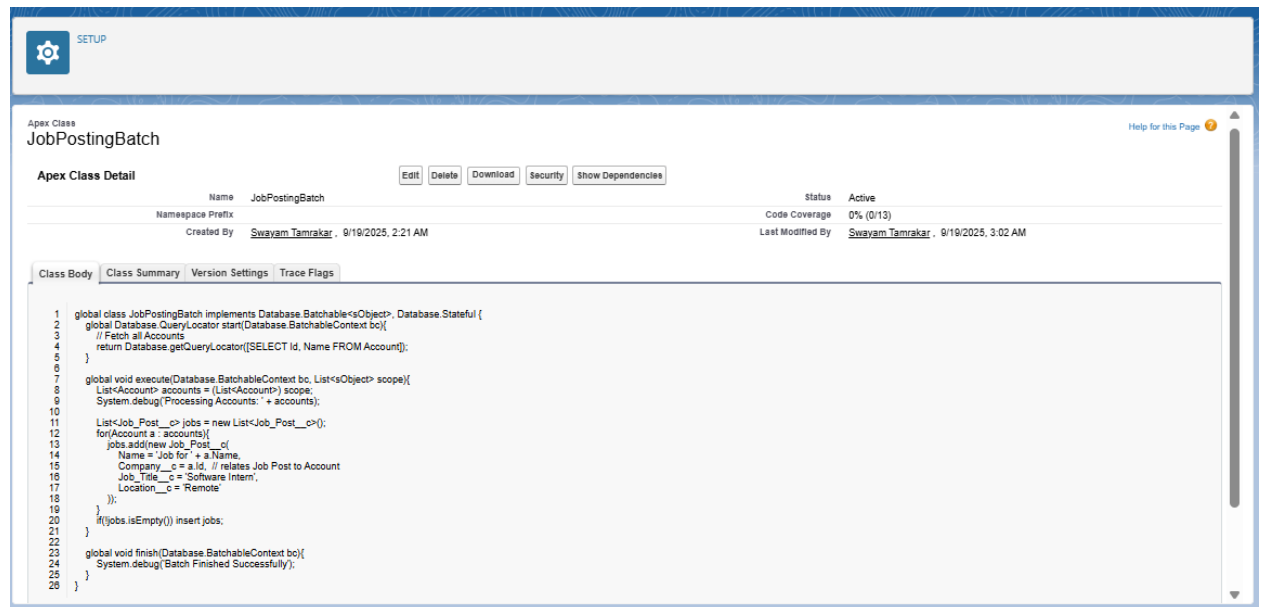
```

        Job_Title__c = 'Software Intern',
        Location__c = 'Remote'
    ));
}

if(!jobs.isEmpty()) insert jobs;
}

global void finish(Database.BatchableContext bc){
    System.debug('Batch Finished Successfully');
}
}

```



2. Queueable Apex

- **Code:**

```

public class ReadinessQueueable implements Queueable {
    private Id studentId;

    public ReadinessQueueable(Id studentId){ this.studentId = studentId; }

    public void execute(QueueableContext context){
        Contact stu = [SELECT Id, GPA__c FROM Contact WHERE Id = :studentId LI
MIT 1];

        Integer skills = [SELECT COUNT() FROM Student_Skill__c WHERE Student
__c = :studentId];
    }
}

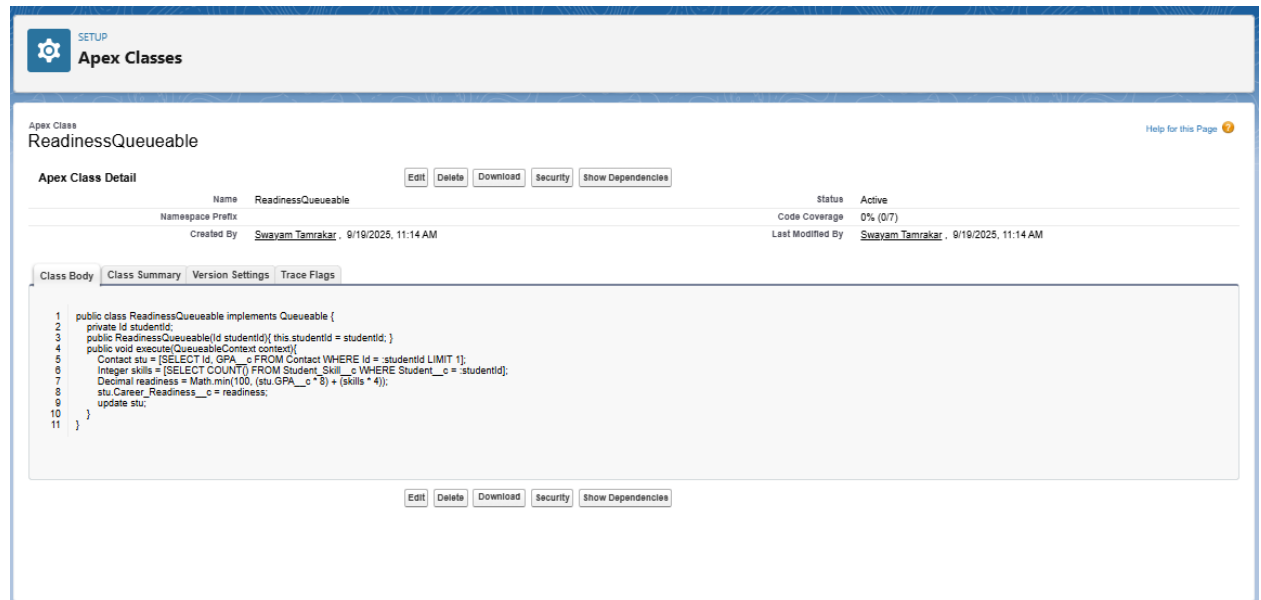
```



```

        Decimal readiness = Math.min(100, (stu.GPA__c * 8) + (skills * 4));
        stu.Career_Readiness__c = readiness;
        update stu;
    }
}

```



3. Scheduled Apex

- **Code:**

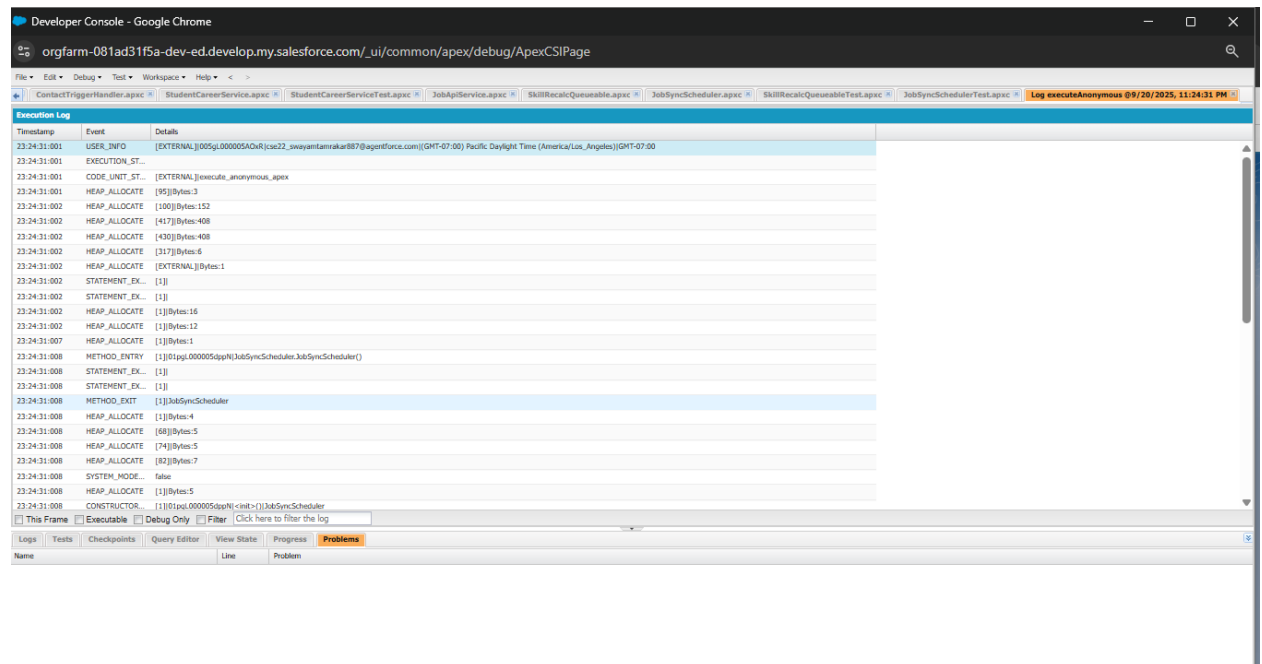
```

global class JobSyncScheduler implements Schedulable {
    global void execute(SchedulableContext sc){
        Database.executeBatch(new JobPostingBatch(), 100);
    }
}

```

- Run in **Execute Anonymous** (Developer Console → Debug → Open Execute Anonymous):

```
System.schedule('Nightly Job Sync', '0 0 23 * * ?', new JobSyncScheduler());
```



4. Future Method

- **Code:**

```
public class StudentNotifier {

    @future(callout=false)

    public static void sendPlacementNotification(Id studentId){

        Contact stu = [SELECT Id, Email, LastName FROM Contact WHERE Id = :studentId LIMIT 1];

        Messaging.SingleEmailMessage mail = new Messaging.SingleEmailMessage();
        mail.setToAddresses(new String[] {stu.Email});

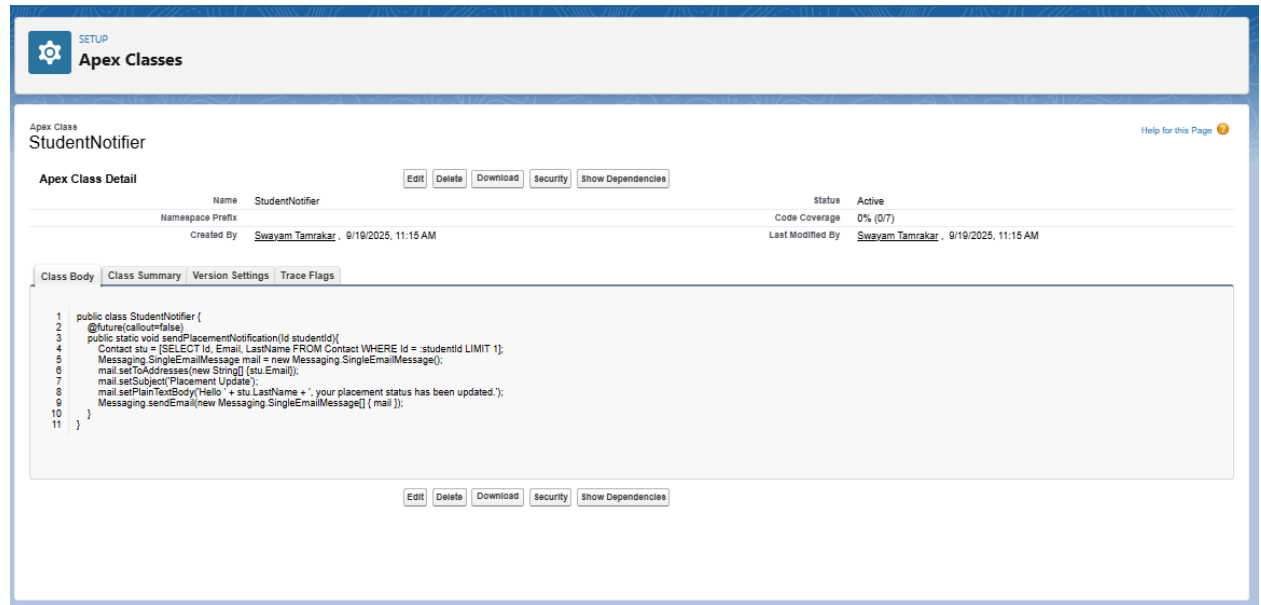
        mail.setSubject('Placement Update');

        mail.setPlainTextBody('Hello ' + stu.LastName + ', your placement status has been updated.');
```

```
        Messaging.sendEmail(new Messaging.SingleEmailMessage[] { mail });

    }

}
```



The screenshot shows the Salesforce Apex Class Editor for a class named **StudentNotifier**. The interface includes a header with the 'Apex Classes' logo and a 'SETUP' link. Below the class name, there are buttons for 'Edit', 'Delete', 'Download', 'Security', and 'Show Dependencies'. A table provides details about the class:

Apex Class Detail		Status
Name	StudentNotifier	Active
Namespace Prefix		Code Coverage 0% (0/7)
Created By	Swayam Tamrakar, 9/19/2025, 11:15 AM	Last Modified By Swayam Tamrakar, 9/19/2025, 11:15 AM

Below the table, there are tabs for 'Class Body', 'Class Summary', 'Version Settings', and 'Trace Flags'. The 'Class Body' tab is selected, showing the following Apex code:

```

1 public class StudentNotifier {
2     @future(callout=false)
3     public static void sendPlacementNotification(Id studentId){
4         Contact stu = [SELECT Id, Email, LastName FROM Contact WHERE Id = :studentId LIMIT 1];
5         Messaging.SingleEmailMessage mail = new Messaging.SingleEmailMessage();
6         mail.setToAddresses(new String[] {stu.Email});
7         mail.setSubject('Placement Update');
8         mail.setPlainTextBody('Hello ' + stu.LastName + ', your placement status has been updated. ');
9         Messaging.sendEmail(new Messaging.SingleEmailMessage[] { mail });
10    }
11 }

```

At the bottom of the class body, there are buttons for 'Edit', 'Delete', 'Download', 'Security', and 'Show Dependencies'.

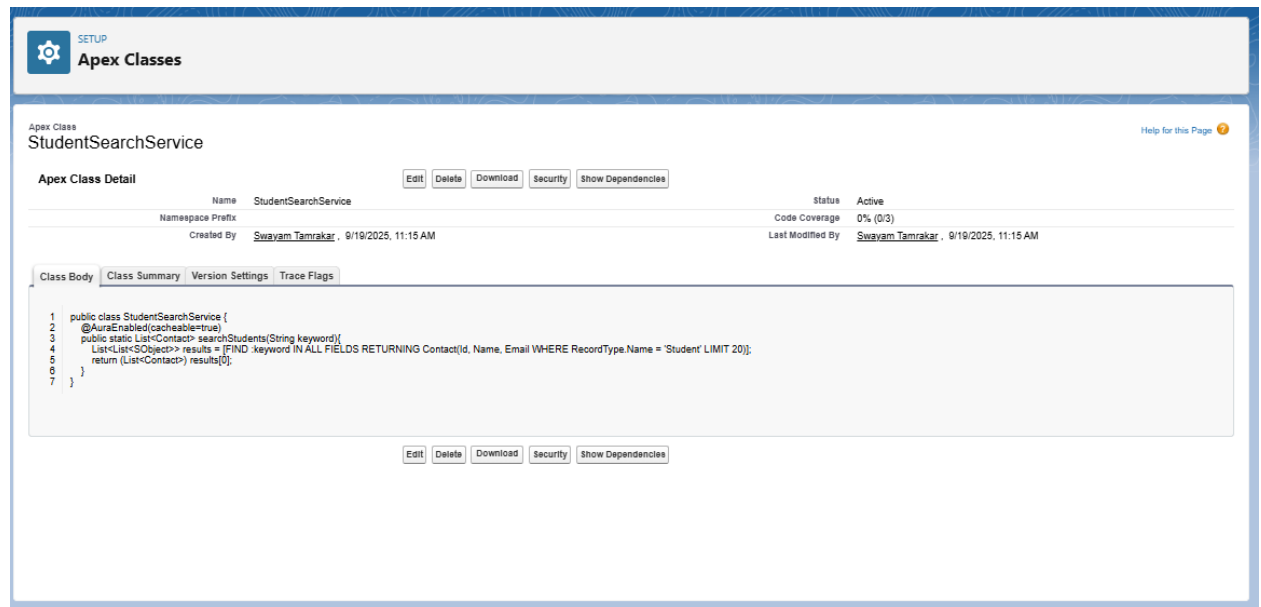
5. SOSL

- **Code:**

```

public class StudentSearchService {
    @AuraEnabled(cacheable=true)
    public static List<Contact> searchStudents(String keyword){
        List<List<SObject>> results = [FIND :keyword IN ALL FIELDS RETURNING Contact(Id
, Name, Email WHERE RecordType.Name = 'Student' LIMIT 20)];
        return (List<Contact>) results[0];
    }
}

```



The screenshot shows the Salesforce Apex Class Editor for a class named **StudentSearchService**. The interface is similar to the previous one, with the 'Apex Classes' logo and 'SETUP' link. Below the class name, there are buttons for 'Edit', 'Delete', 'Download', 'Security', and 'Show Dependencies'. A table provides details about the class:

Apex Class Detail		Status
Name	StudentSearchService	Active
Namespace Prefix		Code Coverage 0% (0/3)
Created By	Swayam Tamrakar, 9/19/2025, 11:15 AM	Last Modified By Swayam Tamrakar, 9/19/2025, 11:15 AM

Below the table, there are tabs for 'Class Body', 'Class Summary', 'Version Settings', and 'Trace Flags'. The 'Class Body' tab is selected, showing the following Apex code:

```

1 public class StudentSearchService {
2     @AuraEnabled(cacheable=true)
3     public static List<Contact> searchStudents(String keyword){
4         List<List<SObject>> results = [FIND :keyword IN ALL FIELDS RETURNING Contact(Id, Name, Email WHERE RecordType.Name = 'Student' LIMIT 20)];
5         return (List<Contact>) results[0];
6     }
7 }

```

At the bottom of the class body, there are buttons for 'Edit', 'Delete', 'Download', 'Security', and 'Show Dependencies'.

6. SOSL

- **Code:**

```
public class SafeUpdateService {  
    public static void safeUpdate(List<Contact> students){  
        try {  
            update students;  
        } catch(DmlException dmlEx){  
            System.debug('DML failed: ' + dmlEx.getMessage());  
        } catch(Exception e){  
            System.debug('Unexpected error: ' + e.getMessage());  
        }  
    }  
}
```

The screenshot displays the Salesforce Apex Classes interface. At the top, there's a 'SETUP' button and the 'Apex Classes' header. Below this, the 'Apex Class' section shows the class name 'SafeUpdateService'. A table lists the class details, including its name, namespace prefix, status (Active), and code coverage (0% (0/5)). The 'Class Body' tab is selected, showing the source code of the class. The code is as follows:

```
1 public class SafeUpdateService {  
2     public static void safeUpdate(List<Contact> students){  
3         try {  
4             update students;  
5         } catch(DmlException dmlEx){  
6             System.debug('DML failed: ' + dmlEx.getMessage());  
7         } catch(Exception e){  
8             System.debug('Unexpected error: ' + e.getMessage());  
9         }  
10    }  
11 }
```

7. JobApiService

- **Code:**

```
public with sharing class JobApiService {
    @future(callout=true)
    public static void fetchJobsAsync(){
        HttpRequest req = new HttpRequest();
        req.setEndpoint('callout:MyJobAPI/v1/jobs'); // Requires Named Credential:
        MyJobAPI
        req.setMethod('GET');
        Http http = new Http();
        HttpResponse res = http.send(req)
        if(res.getStatusCode() == 200){
            List<Object> jobs = (List<Object>)
            JSON.deserializeUntyped(res.getBody());
            List<Job_Post__c> inserts = new List<Job_Post__c>();
            for(Object o : jobs){
                Map<String,Object> jobMap = (Map<String,Object>) o;
                inserts.add(new Job_Post__c(
                    // ✅ Required Fields
                    Name = (String) jobMap.get('title'),           // Record Name
                    Job_Title__c = (String) jobMap.get('title'),    // Required
                    Job_Post__c = jobMap.containsKey('id')
                        ? 'API-' + String.valueOf(jobMap.get('id'))
                        : 'API-' + String.valueOf(Crypto.getRandomInteger()), // Required
                    unique identifier
                    Deadline__c = Date.today().addDays(30),        // Required field
                    // ✅ Optional Fields
                    Title__c = (String) jobMap.get('title'),
                    Location__c = jobMap.containsKey('location') ? (String)
                    jobMap.get('location') : null
                ));
            }
        }
    }
}
```

```

    }
    if(!inserts.isEmpty()) insert inserts;
  }
}
}

```

The screenshot shows the Salesforce Apex Classes page for the class **JobApiService**. The page includes a header with the Salesforce logo and a "SETUP" button. Below the header, the class name "JobApiService" is displayed. A "Help for this Page" link is visible in the top right corner. The "Apex Class Detail" section shows the class name, namespace prefix, and creation details. The "Class Body" tab is selected, displaying the Apex code for the class. The code includes a static method `fetchJobsAsync()` that makes an HTTP GET request to a specified endpoint and returns a list of job objects. The code also includes comments for required and optional fields.

8. Run Tests

- Setup → Apex Test Execution → Select StudentCareerServiceTest → Run.
- Expected: Pass with $\geq 75\%$ coverage.

The screenshot shows the Salesforce Apex Test Execution page. The page includes a header with the Salesforce logo and a "SETUP" button. Below the header, the page title "Apex Test Execution" is displayed. A "Help for this Page" link is visible in the top right corner. The "Select Tests..." button is highlighted. Below the button, a table shows the results of the test execution. The table has columns for Status, Class, and Result. The table shows that the test `StudentCareerServiceTest` was run successfully, with a result of "2/2 Test Methods Passed".

Status	Class	Result
Test Run: 2025-09-20 11:28:52, cse22_swayamtamrakar887@agentforce.com, (1 test class run)		
✓	StudentCareerServiceTest	(2/2) Test Methods Passed
Test Run: 2025-09-20 11:25:18, cse22_swayamtamrakar887@agentforce.com, (1 test class run)		
✗	StudentCareerServiceTest	(1/2) Test Methods Passed



SETUP

Apex Test Execution

Apex Test Result

Help for this Page

Apex Test Result Detail

Time Started	9/20/2025, 11:28 AM
Class	StudentCareerServiceTest
Method Name	testCalloutWithMockAndFutureMethod
Pass/Fail	Pass
Error Message	
Stack Trace	

Phase 6: User Interface Development

6.1 Create Apex controller

1. In the org (Setup) — quick option:

- Setup → Quick Find → **Apex Classes** → **New** → write a code → **Save**.

Apex

```
public with sharing class SkillController {

    @AuraEnabled(cacheable=true)

    public static List<StudentSkillIDTO> getSkillsForStudent(Id studentId){

        if(studentId == null) return new List<StudentSkillIDTO>();

        List<Student__Skill__c> recs = [

            SELECT Id, Proficiency__c,

                Skill__c, Skill__r.Name, Skill__r.Skill_Level__c

            FROM Student__Skill__c

            WHERE Student__c = :studentId

            ORDER BY Skill__r.Name

        ];

        List<StudentSkillIDTO> out = new List<StudentSkillIDTO>();

        for(Student__Skill__c s : recs){

            String skillName = (s.Skill__r != null) ? s.Skill__r.Name : null;

            String skillLevel = (s.Skill__r != null) ? s.Skill__r.Skill_Level__c : null;

            out.add(new StudentSkillIDTO(s.Skill__c, skillName, skillLevel, s.Proficiency__c));

        }

        return out;

    }

    public class StudentSkillIDTO {

        @AuraEnabled public Id skillId;

        @AuraEnabled public String skillName;
```



```

    @AuraEnabled public String skillLevel;

    @AuraEnabled public String proficiency;

    public StudentSkillDTO(Id skillId, String skillName, String skillLevel, String proficiency)
    {
        this.skillId = skillId;

        this.skillName = skillName;

        this.skillLevel = skillLevel;

        this.proficiency = proficiency;
    }
}
}
}

```

The screenshot displays the Salesforce Apex Classes interface. At the top, there's a 'SETUP' button and the 'Apex Classes' header. Below this, the 'Apex Class' section shows 'SkillController'. A 'Help for this Page' link is visible. The 'Apex Class Detail' section includes buttons for 'Edit', 'Delete', 'Download', 'Security', and 'Show Dependencies'. A table lists the class details:

Name	SkillController	Status	Active
Namespace Prefix		Code Coverage	0% (0/15)
Created By	Swayam Tamrakar , 9/21/2025, 9:59 AM		Last Modified By
			Swayam Tamrakar , 9/21/2025, 9:59 AM

Below the table, there are tabs for 'Class Body', 'Class Summary', 'Version Settings', and 'Trace Flags'. The 'Class Body' tab is selected, showing the following Apex code:

```

1 public with sharing class SkillController {
2     @AuraEnabled(cacheable=true)
3     public static List<StudentSkillDTO> getSkillsForStudent(Id studentId){
4         if(studentId == null) return new List<StudentSkillDTO>();
5
6         List<Student_Skill__c> recs = [
7             SELECT Id, Proficiency__c,
8                 Skill__c, Skill__r.Name, Skill__r.Skill_Level__c
9             FROM Student_Skill__c
10            WHERE Student__c = :studentId
11            ORDER BY Skill__r.Name
12        ];
13
14        List<StudentSkillDTO> out = new List<StudentSkillDTO>();
15        for(Student_Skill__c s : recs){

```

Notes:

- This returns a small DTO list ready for LWC.
- cacheable=true lets the wire adapter cache results for performance.

6.2 Create the LWC (skillTracker) in VS Code

1. In VS Code

- Command palette → **SFDX: Create Project**.
- Connect to org: **SFDX: Authorize an Org** → choose your Developer Org.
- In the force-app/main/default/lwc folder → right-click → **SFDX: Create Lightning Web Component** → name it skillTracker.

You'll get three files. Replace them with the code below.

- **skillTracker.html**

```
<template>
```

```
<lightning-card title="Skills" icon-name="standard:skills">
```

```
<template if:true={skills.data}>
```

```
<div class="slds-p-around_small">
```

```
<template for:each={skills.data} for:item="s">
```

```
<div key={s.skillId} class="skill-row slds-grid slds-grid_align-spread slds-p-vertical_xx-small">
```

```
<div class="slds-col">{s.skillName}</div>
```

```
<div class="slds-col slds-text-color_weak">{s.skillLevel}</div>
```

```
<div class="slds-col slds-text-color_weak">{s.proficiency}</div>
```

```
</div>
```

```
</template>
```

```
</div>
```

```
</template>
```

```
<template if:true={skills.error}>
```

```
<div class="slds-p-around_small slds-text-color_error">
```

```
Error loading skills.
```

```
</div>
```

```
</template>
```

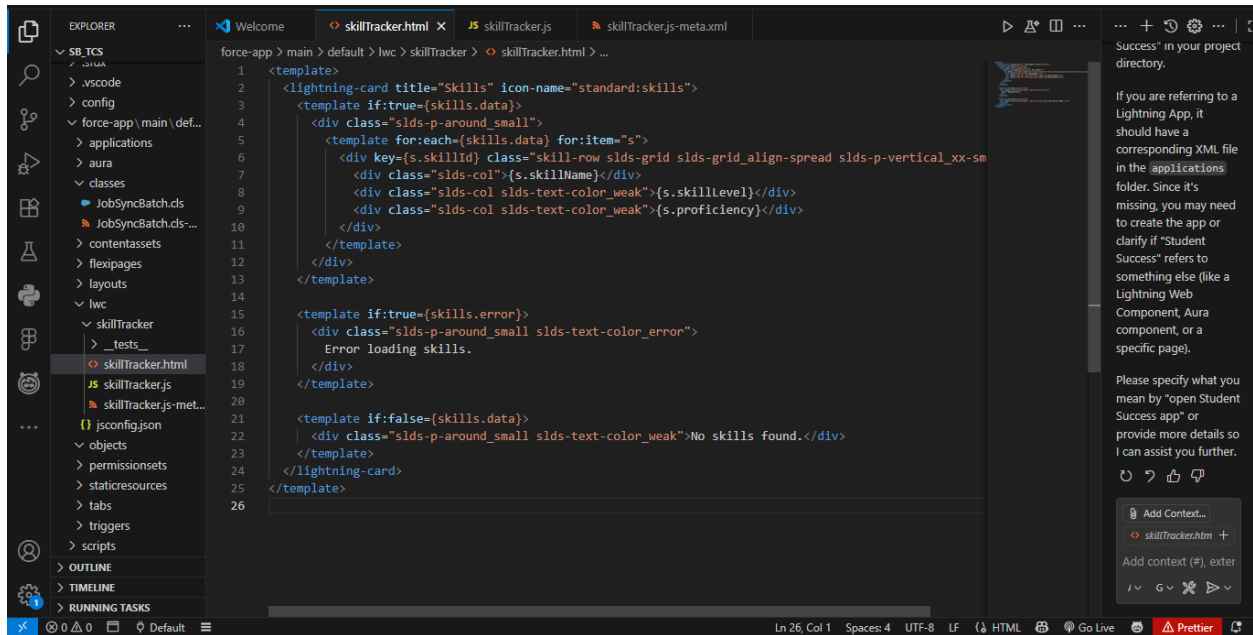
```
<template if:false={skills.data}>
```

```
<div class="slds-p-around_small slds-text-color_weak">No skills found.</div>
```

```
</template>
```

```
</lightning-card>
```

```
</template>
```



- **skillTracker.js**

```
import { LightningElement, api, wire } from 'lwc';
```

```
import getSkillsForStudent from '@salesforce/apex/SkillController.getSkillsForStudent';
```

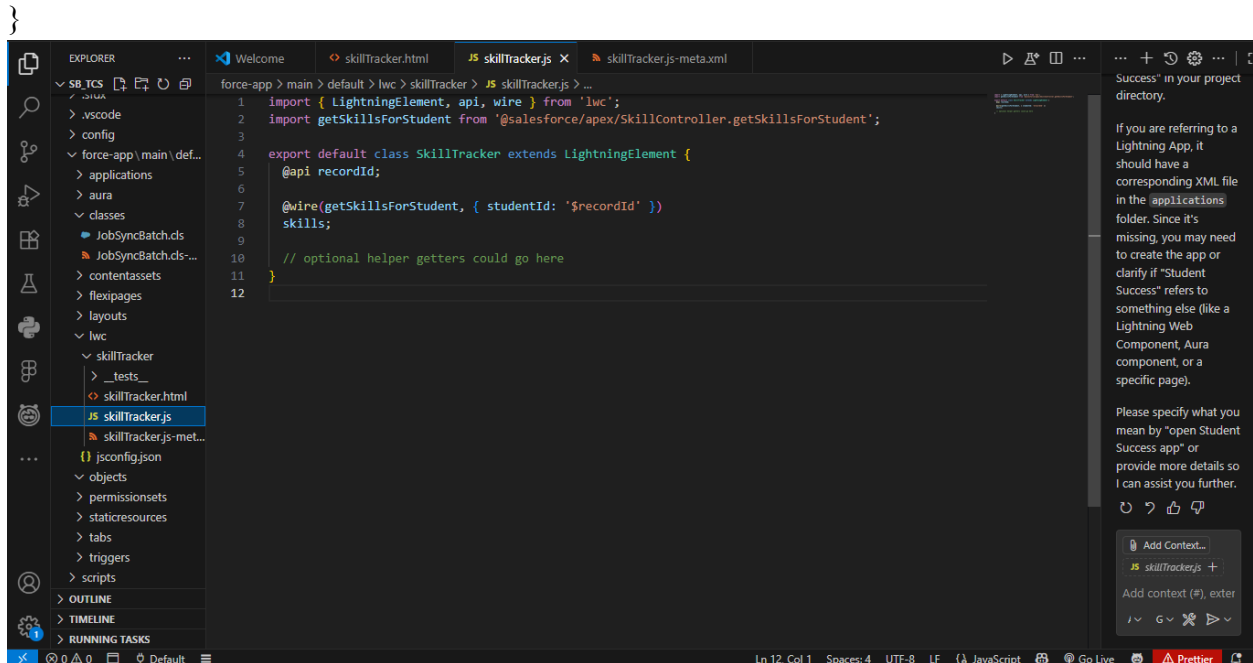
```
export default class SkillTracker extends LightningElement {
```

```
  @api recordId;
```

```
  @wire(getSkillsForStudent, { studentId: '$recordId' })
```

```
  skills;
```

```
  // optional helper getters could go here
```



- **skillTracker.js-meta.xml**

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
```

```
  <apiVersion>59.0</apiVersion>
```

```
  <isExposed>true</isExposed>
```

```
  <targets>
```

```
    <target>lightning__RecordPage</target>
```

```
    <target>lightning__AppPage</target>
```

```
  </targets>
```

```
  <targetConfigs>
```

```
    <targetConfig targets="lightning__RecordPage">
```

```
      <objects>
```

```
        <object>Contact</object>
```

```
      </objects>
```

```
    </targetConfig>
```

</targetConfigs>

</LightningComponentBundle>

2. Deploy to Org

- In VS Code explorer → right-click component folder → **SFDX: Deploy Source to Org**.
- Or run CLI: `sfdx force:source:deploy -p force-app/main/default/lwc/skillTracker`.

6.3 Add the LWC & Components to Contact Record Page

1. Create Lightning App

- Setup → **App Manager** → **New Lightning App**
 - App Name: Student Success → Next → choose branding (color/logo) → Next.
 - Add Navigation Items: **Contacts**, **Internships** (your custom object), **Placements**, **Job_Posts**, **Reports**, **Dashboards**, **Student Skills**, **Skills**, e.t.c → Finish.

App Settings

- App Details & Branding**
- App Options
- Utility Items (Desktop Only)
- Navigation Items
- User Profiles

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ⓘ
Student Success

* Developer Name ⓘ
Student_Success

Description ⓘ
An app to track student performance, internships, placements, and provide AI-...

App Branding

Image ⓘ


Primary Color Hex Value ⓘ
 #FAFE01

Clear

Org Theme Options
☐ Use the app's image and color instead of the org's custom theme

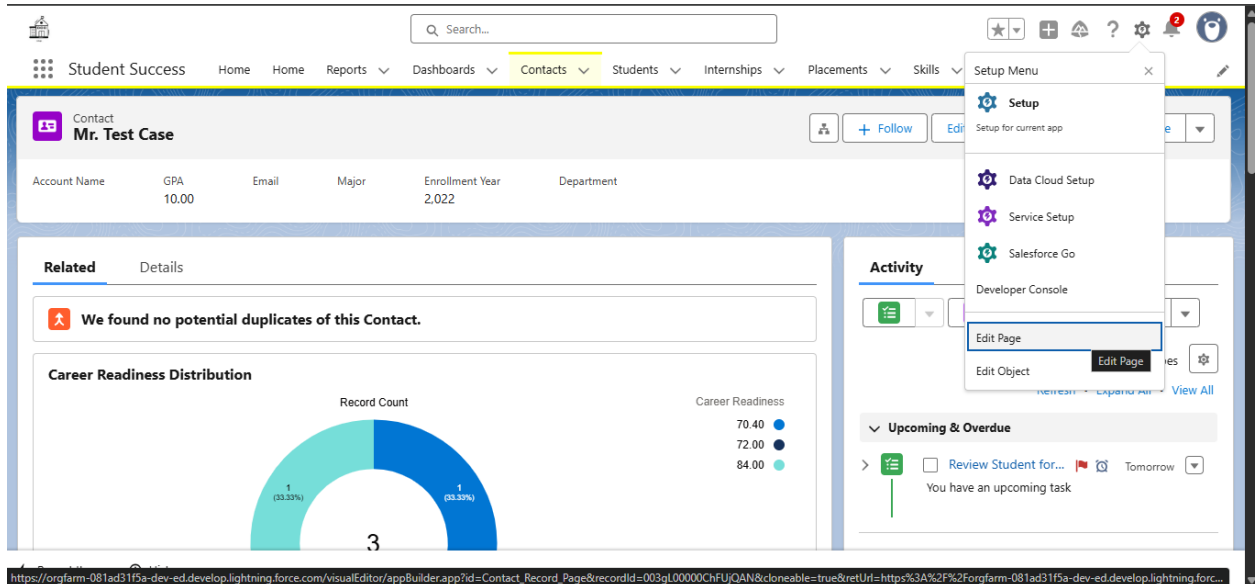
App Launcher Preview



Student Success
An app to track student performance, internships, plac...

2. Edit Contact Record Page

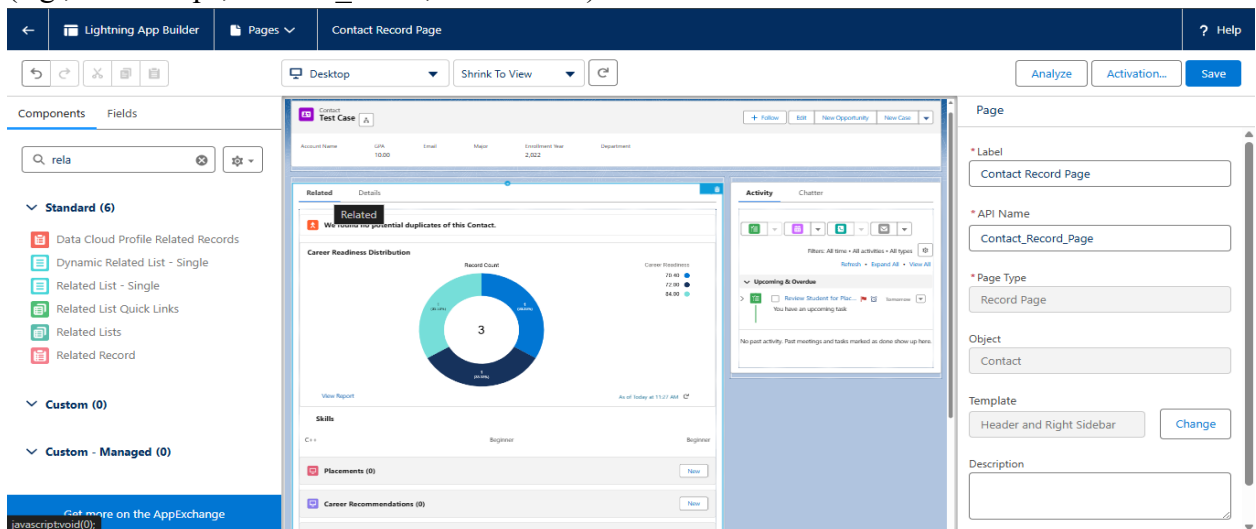
- App Launcher → open **Contacts** → open any Contact record → click the gear icon (⚙️) top-right → **Edit Page**. This opens **Lightning App Builder**.



- You're now editing the Contact Record Page. Do these:

Add Related Lists

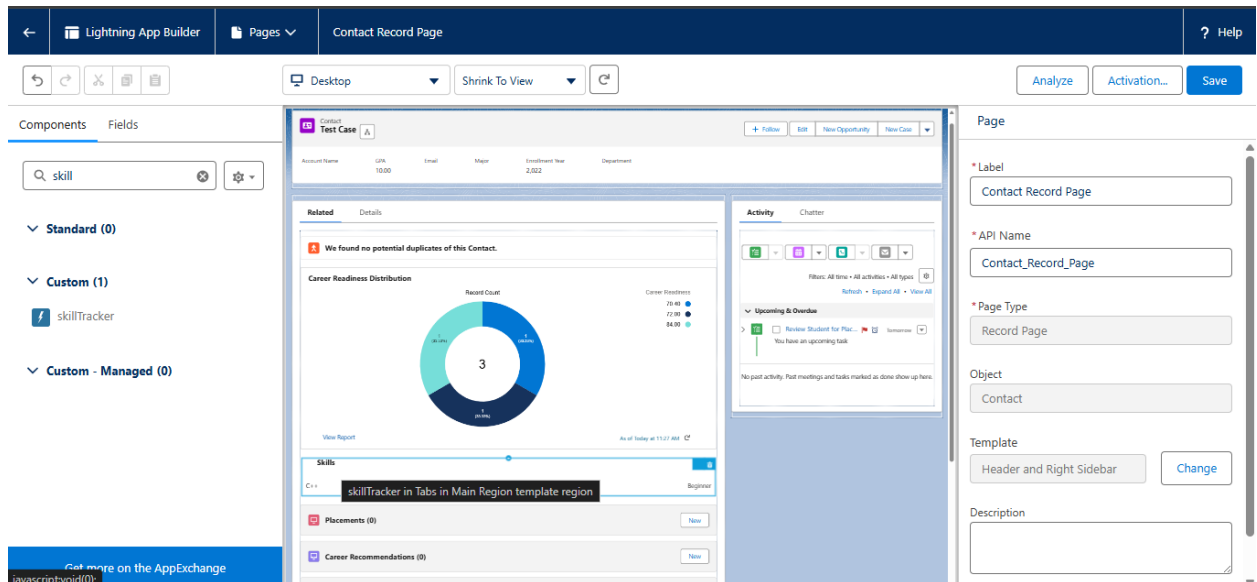
- From left components panel → drag **Related Lists** (or **Related List - Single**) onto the page where you want them.
- Click the related list component → in the right pane choose which related list to show (e.g., Internships, Student_Skills, Placements).



- Save & Activate.

Add the skillTracker LWC

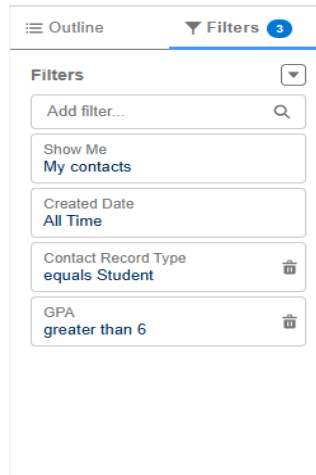
- In left panel → find **Custom** section → your skillTracker component → drag it onto the page (a column/region).



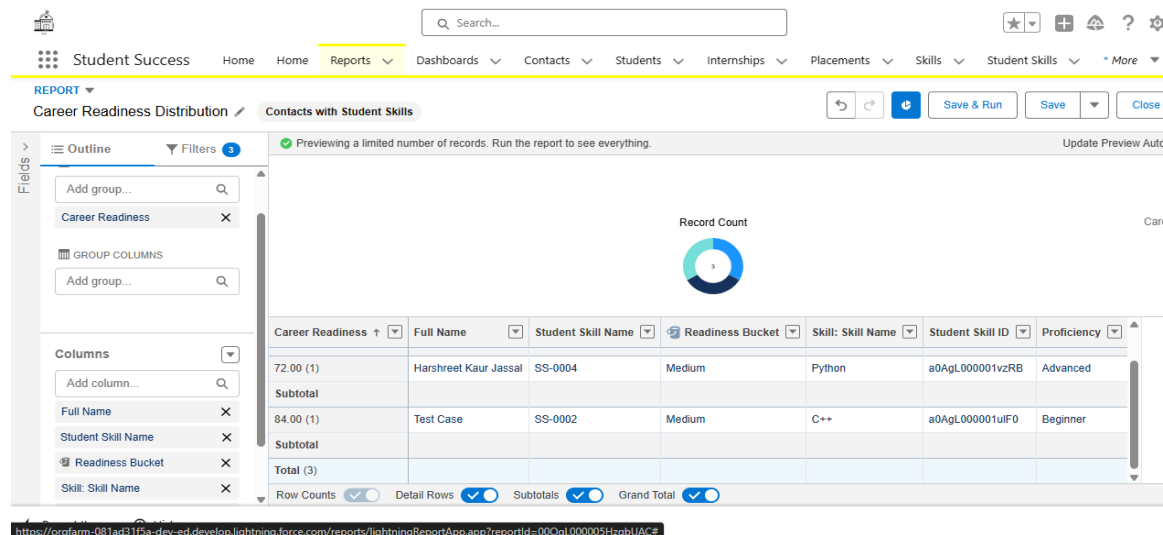
Add a Report Chart (Career Readiness distribution)

- Create a report.
 - Create a Custom Report Type for Student Skills
 - Go to **Setup**.
 - Quick Find → **Report Types** → click **New Custom Report Type**.
 - Configure:
 - ✓ **Primary Object:** Contacts
 - ✓ **Report Type Label:** Contacts with Student Skills
 - ✓ **Description:** Report showing Contacts and their related Student Skills.
 - ✓ **Category:** Choose Other Reports
 - ✓ **Deployment Status:** Deployed
 - Click **Next**.
 - Related Object: Student_Skill__c.
 - Choose: “Each A record must have at least one related B record.”
 - **Save**
 - Create a Report from this New Report Type
 - App Launcher → **Reports** → **New Report**.
 - In Report Type selector, search for **Contacts with Student Skills**.
 - Add Fields:
 - ✓ Name

- ✓ Career_Readiness__c
 - ✓ Skill__r.Name
 - ✓ Proficiency__c
- Buckets:
 - ✓ In the left **Fields** pane, find the Career_Readiness__c field.
 - ✓ Click the chevron (▼) next to that field and choose **Bucket this field** (or click **Add | Bucket Field** at top of Fields pane).
 - ✓ Give the bucket a name: **Readiness Bucket**.
 - ✓ Choose **Bucket By: Numeric Range** (or the UI will allow you to create ranges).
 - ✓ Create ranges, for example:
 - Low : 0 to 30
 - Medium : 31 to 60
 - High : 61 to 100
 - Group rows:
 - ✓ Drag the **Readiness Bucket** from **Fields** into the **Group Rows** area on the left (where it says *Add group...*).
 - ✓ You should now see **GROUP ROWS** populated — the report preview updates.
 - Add Filter:
 - ✓ On the right side, open the **Filters** pane.
 - ✓ By default you may see a Date filter (e.g., Show Me / Created Date). Remove or set it to *All Time*.
 - ✓ Click **Add Filter**.
 - Field: type **Record Type** → choose **Record Type** (Contact.Record Type).
 - Operator: **equals**.
 - Value: choose **Student**
 - Click **Apply**.



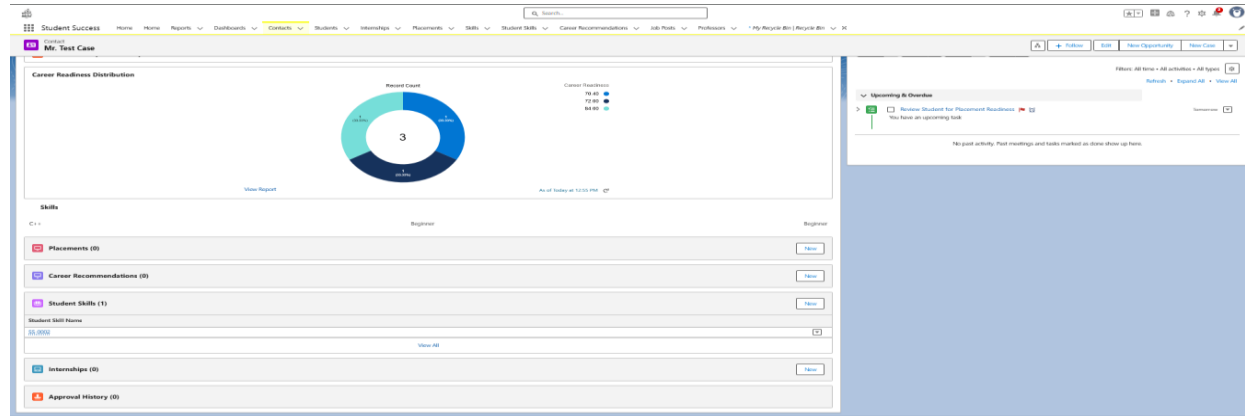
- ✓ Optional: add any other filters (Department, GPA > 6.0, etc).
- Add Chart:
 - ✓ Once you have a group, **Add Chart** becomes active (top bar). Click **Add Chart**.
 - ✓ In the chart configuration panel:
 - **Chart Type**: choose **Donut** (good for distribution) or **Bar**.
 - **Grouping**: should already be set to the group you made
 - **Values**: default is **Record Count**
 - Optional: show percentage or values in chart options.
- Click Done



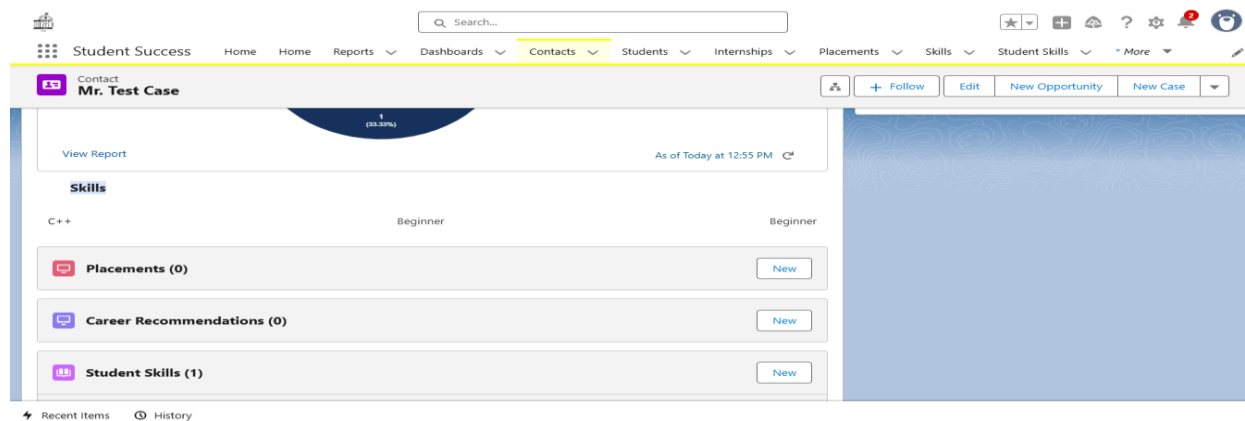
- Save & Run the report.
 - Click **Save & Run**.
 - Name the report: **Career Readiness Distribution**.
 - Choose a folder (e.g., My Personal Reports or a team folder).
 - Click **Save** → then **Run**.

6.4 Test everything in Org

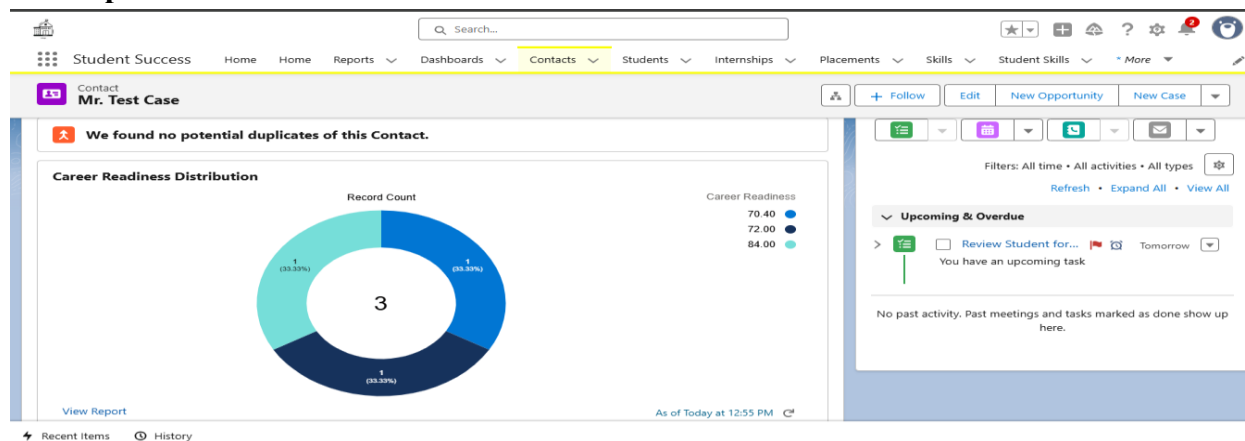
1. Create a **Contact** with Record Type Student. Add GPA and other fields.
2. Create a **Skill__c** record and a **Student_Skill__c** linking the Student.
3. Open the Student Contact record. The page should show:
 - Related lists (Internships, Skills).



- The **Skills** LWC showing skill name, skill level, proficiency.



- The **Report Chart** with readiness distribution.



Phase 7: Integration & External Access

7.1 Named Credentials (preferred for callouts)

When to use: Salesforce makes *callouts to external services* (LMS, Job APIs). **Named Credentials** centralize the endpoint + auth. Use them instead of hard-coding endpoints.

A. If external API is public or uses simple credentials

1. Setup → **Named Credentials** → **New Named Credential**.
2. Fill:
 - **Label:** LMS_NamedCred
 - **Name:** LMS_NamedCred
 - **URL:** https://api.mylms.com (replace with actual API root)
 - **Identity Type:** **Named Principal** (single identity for all callouts)
 - **Authentication Protocol:** **Password Authentication** (or **No Authentication** for public)
 - Enter username/password if using password auth.

3. Save.

The screenshot shows the 'LMS_NamedCred' configuration page in Salesforce Setup. The breadcrumb trail is 'SETUP > NAMED CREDENTIALS'. The page title is 'LMS_NamedCred'. There are 'Edit' and 'Delete' buttons in the top right corner. The form contains the following fields:

- Label:** LMS_NamedCred
- Name:** LMS_NamedCred
- URL:** https://github.com/settings/personal-access-tokens/8775376
- Enabled for Callouts:** A toggle switch that is currently turned on (checked).
- Authentication:** A section with two options: 'External Credential' and 'No Authentication' (which is selected).
- Client Certificate:** A field for uploading a certificate.
- Callout Options:** A section with four checkboxes:
 - Generate Authorization Header:** Checked.
 - Allow Formulas in HTTP Header:** Not checked.
 - Allow Formulas in HTTP Body:** Not checked.
 - Outbound Network Connection:** Not checked.
- Managed Package Access:** A section with two fields:
 - Created By Namespace:** A field with a help icon.
 - Allowed Namespaces for Callouts:** A field with a help icon.

B. If external API uses OAuth 2.0 (recommended) — create an Auth. Provider first

1. Setup → **Auth. Providers** → **New**.

- **Provider Type:** choose the provider that matches the API (e.g., **GitHub** or the specific vendor if listed).
- **Name:** LMS_AuthProvider
- **Consumer Key / Secret:** get these from the external API developer console (not Salesforce).
- **Authorize Endpoint URL / Token Endpoint URL / User Info Endpoint URL:** paste the external API's OAuth URLs.

- **Default Scopes:** e.g., read write (depends on external API).
- Save.

The screenshot shows the 'Auth. Providers' setup page in Salesforce. The page title is 'Auth. Provider' with a 'Back to List: Custom Apps' link. A 'Help for this Page' icon is in the top right. The 'Auth. Provider Detail' section includes buttons for 'Edit', 'Delete', and 'Clone'. The details are as follows:

Auth. Provider ID	0SOgl.000000rie1
Provider Type	GitHub
Name	LMS_AuthProvider
URL Suffix	LMS_AuthProvider
Consumer Key	https://github.com/settings/personal-access-tokens/8775376
Consumer Secret	Click to reveal
Authorize Endpoint URL	https://github.com/login/oauth/authorize
Token Endpoint URL	https://github.com/login/oauth/access_token
Default Scopes	read
Include Consumer Secret in SOAP API Responses	☒ i
Custom Error URL	
Custom Logout URL	
Portal	
Icon URL	
Use Salesforce MFA for this SSO Provider	☐ i

The 'Salesforce Configuration' section includes the following URLs:

Test-Only Initialization URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/test/LMS_AuthProvider
Existing User Linking URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/link/LMS_AuthProvider
OAuth-Only Initialization URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/oauth/LMS_AuthProvider
Callback URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/callback/LMS_AuthProvider
Single Logout URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/tp/oidc/logout

Buttons for 'Edit', 'Delete', and 'Clone' are located at the bottom of the configuration section.

2. After Save, on the Named Credential detail, click **Authenticate** (or **Authorize**). That opens the external provider's login page. Complete the OAuth flow and the Named Credential will store the access token/refresh token.

Use in Apex (callout):

```
HttpRequest req = new HttpRequest();
```

```
req.setEndpoint('callout:LMS_NamedCred/v1/courses'); // callout: + NamedCredential + path
```

```
req.setMethod('GET');
```

```
Http http = new Http();
```

```
HttpResponse res = http.send(req);
```

```
System.debug(res.getStatus());
```

```
System.debug(res.getBody());
```

Notes:

- Use callout:NamedCredentialName/path as the endpoint — Salesforce replaces host & auth.
- Named Credentials handle token refresh for OAuth providers automatically.

7.2 Platform Events

When to use: publish real-time notifications to external systems or internal subscribers (e.g., notify recruiter system when a student is shortlisted).

Create the Platform Event

1. Setup → **Platform Events** → **New Platform Event**.
 - **Label:** StudentShortlisted
 - **Object Name:** StudentShortlisted__e (platform events end with __e)
 - **Publish Behavior:** **Publish After Commit** (recommended)
 - Save.

The screenshot shows the Salesforce 'Platform Events' setup page for an event named 'StudentShortlisted'. The page has a blue header with the 'SETUP Platform Events' logo. Below the header, the event name 'StudentShortlisted' is displayed. There are links for 'Standard Fields (4)' and 'Custom Fields & Relationships (4)'. The main section is titled 'Platform Event Definition Detail' and contains a table with the following information:

Platform Event Definition Detail		Edit	Delete
Singular Label	StudentShortlisted		
Plural Label	StudentShortlisteds		
Object Name	StudentShortlisted		
API Name	StudentShortlisted__e		
Event Type	High Volume		
Publish Behavior	Publish After Commit		
Created By	Swayam Tamrakar, 9/23/2025, 2:49 AM		
Modified By	Swayam Tamrakar, 9/23/2025, 2:49 AM		

2. Add fields: while on the event detail → **Fields & Relationships** → **New**:
 - **Field 1:** StudentId__c — Data Type: Text (Length: 18)

- **Field 2:** JobId__c — Data Type: Text (Length: 18)
- (Add any other fields like ShortlistReason__c, ShortlistedBy__c)

The screenshot shows the 'Platform Events' setup page in Salesforce. The main section is 'Platform Event Definition Detail' for the event 'StudentShortlisted__e'. It includes fields for Singular Label, Plural Label, Object Name, API Name, Event Type, Publish Behavior, and Created/Modified By. Below this are three tables: 'Standard Fields', 'Custom Fields & Relationships', and 'Triggers'.

Action	Field Label	Field Name	Data Type	Controlling Field	Indexed
Edit Del	Created By	CreatedBy	Lookup(User)		
Edit Del	Created Date	CreatedDate	Date/Time		
Edit Del	Event UUID	EventUuid	Text(36)		
Edit Del	Replay ID	ReplayId	External Lookup		

Action	Field Label	API Name	Data Type	Indexed	Controlling Field	Modified By
Edit Del	JobId	JobId__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:51 AM
Edit Del	ShortlistedBy	ShortlistedBy__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:51 AM
Edit Del	ShortlistReason	ShortlistReason__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:51 AM
Edit Del	StudentId	StudentId__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:50 AM

Action	Name	Api Version	Status	Size Without Comments	Last Modified By
Edit Del	StudentShortlistedTrigger	64.0	Active	493	Swayam Tamrakar, 9/23/2025, 2:55 AM

Action	Subscriber	Last Processed Id	Last Published Id	State	Batch Size	User
Manage	StudentShortlistedTrigger	19527	Not Available	Running	2,000	Automated Process

Publish a Platform Event (Apex example)

```
StudentShortlisted__e evt = new StudentShortlisted__e(
    StudentId__c = '0033t00002ABCDE',
    JobId__c = 'a0E3t00000XYZ12'
);
// Publish
```

```
Database.SaveResult sr = EventBus.publish(evt);
```

```
System.debug('Published: ' + sr.isSuccess());
```

Subscribe to Platform Events

- **Apex trigger subscriber** (internal processing):

```
trigger StudentShortlistedTrigger on StudentShortlisted__e (after insert) {
    for (StudentShortlisted__e evt : Trigger.New) {
```

```
// e.g., create a Task for Placement Officer

// evt.StudentId__c, evt.JobId__c

}

}
```

Test: use Execute Anonymous to run the Apex publish code. Check subscriber behavior (e.g., a Trigger that creates a Task).

7.3 Remote Site Settings (when Named Credentials not used)

When to use: quick test callouts or older patterns — *less secure* than Named Credentials.

1. Setup → **Remote Site Settings** → **New Remote Site**.
 - **Remote Site Name:** mylms_api
 - **Remote Site URL:** https://api.mylms.com
 - **Description:** Allow callouts to LMS API
 - Save.

Remote Site Settings

Remote Site Details [Back to List: Custom Apps](#) [Help for this Page](#)

Remote Site Detail [Edit](#) [Delete](#) [Clone](#)

Remote Site Name	mylms_api	Modified By	Swayam Tamrakar, 9/23/2025, 2:59 AM
Remote Site URL	https://api.mylms.com		
Disable Protocol Security	☐		
Description	Allow callouts to LMS API		
Active	☒		
Created By	Swayam Tamrakar, 9/23/2025, 2:59 AM		

[Edit](#) [Delete](#) [Clone](#)

Use: In Apex callouts without Named Credential, you use the full https URL:

```
HttpRequest req = new HttpRequest();
```



```
req.setEndpoint('https://api.mylms.com/v1/courses');
```

```
req.setMethod('GET');
```

```
Http http = new Http();
```

```
HttpResponse res = http.send(req);
```

Recommendation: prefer **Named Credentials** — they centralize, secure, and make token refresh automatic.

Quick Testing checklist (do these after setup)

1. **Connected App:** verify Consumer Key & Secret are visible on the Connected App detail page. If you used Admin-approved, add your admin profile under **Manage Profiles**.
2. **Named Credential:** after creating + authenticating, run this sample from **Developer Console** → **Execute Anonymous**:

```
HttpRequest req = new HttpRequest();
```

```
req.setEndpoint('callout:LMS_NamedCred/v1/ping'); // replace with an endpoint that exists
```

```
req.setMethod('GET');
```

```
Http http = new Http();
```

```
HttpResponse res = http.send(req);
```

```
System.debug(res.getStatus() + ' ' + res.getBody());
```

3. **External Service:** build a simple Flow that calls one External Service action and run it.
 4. **Platform Event:** execute the Apex publish snippet. Then check that your Subscriber Trigger executed (e.g., created a Task or wrote to debug log).
 5. **Remote Site:** if using Remote Site, try a plain Apex callout to the external URL.
-

Phase 8: Data Management & Deployment

8.1 Data import — Data Import Wizard

Prepare: Students.csv example headers:

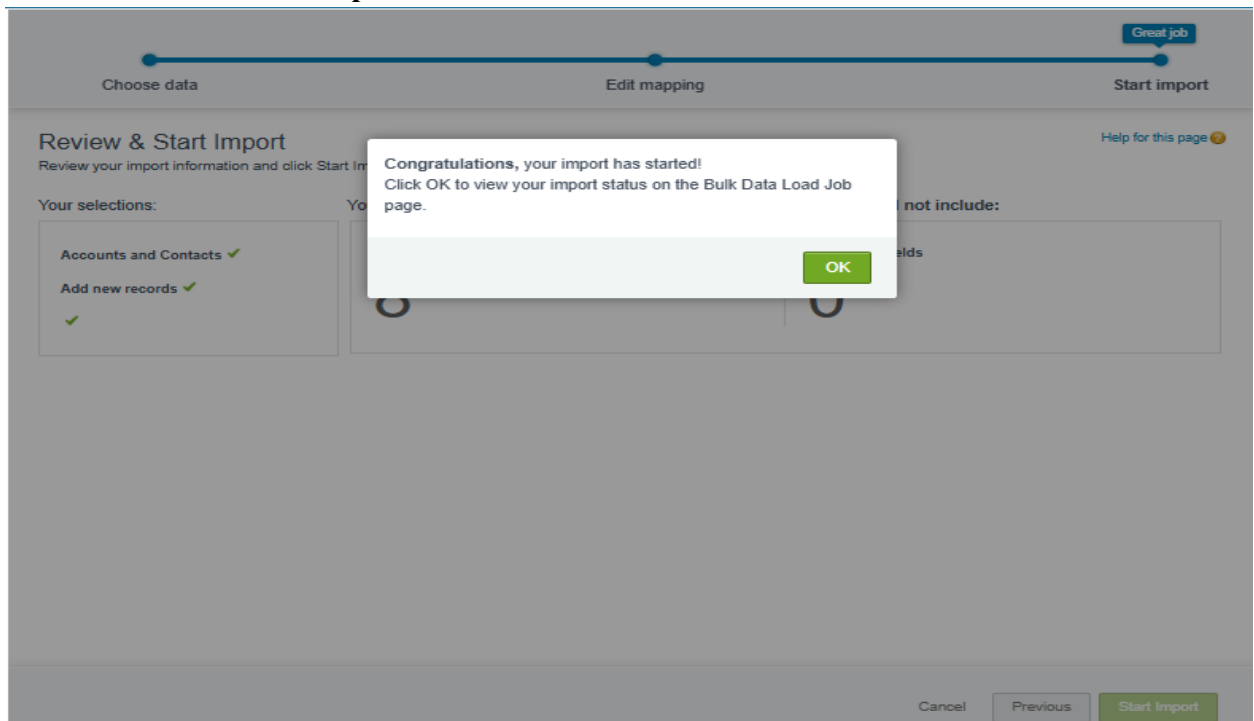
FirstName,LastName,Email,Student_ID__c,AccountName,GPA__c

Swayam,Tamrakar,swayam@example.com,STU001,XYZ Engineering College,8.2

Note: You can import Contacts and specify Account by **Account Name** (Data Import Wizard will try to match or create).

Steps

1. In Salesforce: click  → **Setup**.
2. Quick Find → **Data Import Wizard** → **Launch Wizard**.
3. Choose the object group (e.g., **Standard Objects** → Accounts and Contacts).
4. Select operation: **Add new records** (or Update existing if you include Salesforce IDs).
5. Choose CSV file (Students.csv) and follow the mapping screen: map CSV columns to Salesforce fields.
 - Map AccountName → Account Name (if importing Contacts tied to Account).
6. Review and click **Start Import**.



7. After run completes, check the email notification and import results page for success/error counts.
8. Fix errors (download error CSV), correct CSV, re-import failed rows.

Bulk Data Load Job
750gL00000E24CO

View the details of a bulk data load job.

[Back to List: Bulk Data Load Jobs](#)

Bulk Data Load Job Detail		Job ID		Job Type		Bulk V1		Status		Closed	
Submitted By	Swayam Tarrakar	Operation	Insert	Total Processing Time (ms)	806						
Start Time	9/24/2025, 2:21 AM PST	Queued Batches	0	API Active Processing Time (ms)	601						
End Time	9/24/2025, 2:21 AM PST	In Progress Batches	0	Apex Processing Time (ms)	170						
Time to Complete (hh:mm:ss)	00:01	Completed Batches	1								
Object	Account	Failed Batches	0								
External ID Field		Progress	100%								
Content Type	CSV	Records Processed	100								
Concurrency Mode	Parallel	Records Failed	0								
API Version	64.0	Retries	0								

[Batches](#)

Batch ID	Start Time	End Time	Total Processing Time (ms)	API Active Processing Time (ms)	Apex Processing Time (ms)	Records Processed	Records Failed	Retry Count	State Message	Status
751gL00000BbYSe	9/24/2025, 2:21 AM	9/24/2025, 2:21 AM	806	601	170	100	0	0		Completed

Salesforce import of "Students_100 (1).csv" has finished. 100 rows were processed.

External

Inbox x



noreply@salesforce.c...

14:52 (0 minutes ago)



to me

Your Accounts and Contacts imports are complete. Here are your results:

Accounts Created: 100

Accounts Updated: 0

Accounts Ignored: 0 (We ignored updates that we couldn't match to an existing record.)

Accounts Failed: 0 (We couldn't import these due to errors.)

Processed job information for imported Accounts: <https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/750gL00000E24COQAZ?fromEmail=1>

Tips

- For initial data load, import Accounts first (Colleges/Companies), then Contacts (Students) referencing accounts.
- If you plan to use Data Loader later, you can import basic data with the Wizard and use Data Loader for complex objects.

8.1 Data import — Data Loader (bulk, lookups, upsert)

Best for: bulk loads, custom objects, junction objects, upsert using External IDs.

Install & Login

1. Install Data Loader and open it.
2. Click **Login** → choose **Username-Password** (enter security token if required) OR use **OAuth** flow (recommended).
 - For Developer Org, if using username&password, append your security token to password.

Common operations: Insert, Update, Upsert, Export.

Example: Load Student_Skill__c (junction)

Problem: Student_Skill__c has lookup to Contact and Skill. You likely don't have Salesforce Ids initially.

Use IDs

1. Export Contacts (after Contacts import) to get Contact IDs: Setup → Data Loader → Export → select Contact → export Id, Email, Student_ID__c to CSV.
2. Prepare Student_Skill.csv:

Student__c,Skill__c,Proficiency__c

0033t00002ABCDe, a0H3t00000Skill1,Intermediate

0033t00002FGHIj, a0H3t00000Skill2,Beginner

3. In Data Loader: **Insert** → select Student_Skill__c → choose CSV → Map fields (or auto map) → Finish → check success/error files.

Use External ID upsert (no Salesforce IDs)

1. Create an External ID field on Contact:
 - Setup → Object Manager → Contact → Fields & Relationships → **New** → Text → Student_External_Id__c and check **External ID**.
2. Ensure Skill__c has External ID (or create Skill_Code__c External ID).
3. Prepare Student_Skill_Upsert.csv:

Student_External_Id__c,Skill_Code__c,Proficiency__c

STU001,SKL_JAVA,Intermediate

STU002,SKL_PYTHON,Advanced

4. In Data Loader: choose **Upsert** → select Student_Skill__c → for the lookup fields use the ExternalId mapping: map Student_External_Id__c → Student__r.Student_External_Id__c and Skill_Code__c → Skill__r.Skill_Code__c (Data Loader supports parent external id mapping using relationship notation).
5. Run import and review success/error files.

Tips

- Always **export a small sample** first to confirm mappings.
- Keep the **success.csv** and **error.csv** produced by Data Loader — they tell you row-level outcome and error messages.
- For large loads split into batches < 10k if necessary. Use Bulk API mode in Data Loader for huge data.

8.2 Duplicate Rules — create matching & duplicate rules

Goal: Prevent or alert on duplicate Student (Contact) records by Email or Student_ID__c.

Create Matching Rule

1. Setup → Quick Find → **Matching Rules** → **New Rule**.
 - Select **Contact** as Object.
 - Rule Name: Contact_Match_By_Email_Or_StudentID.
 - Add Matching Criteria:
 - Field: Email Match Method: Exact

- OR add Student_ID__c Match Method: Exact (if you have Student_ID__c).
2. Save and **Activate** the matching rule (must be active to use).

The screenshot shows the 'Matching Rules' setup page in Salesforce. The page title is 'Matching Rules' with a 'd' logo and 'SETUP' text. Below the title, the specific rule is 'Contact_Match_By_Email_Or_StudentID'. There is a 'Help for this Page' link with a question mark icon. A 'Back to List: Custom Apps' link is also present. The 'Matching Rule Detail' section includes buttons for 'Delete', 'Clone', and 'Deactivate'. The details are as follows:

Object	Contact
Rule Name	Contact_Match_By_Email_Or_StudentID
Unique Name	Contact_Match_By_Email_Or_StudentID
Description	
Matching Criteria	Contact: Email EXACT MatchBlank = FALSE
Status	Active
Created By	Swayam Tamrakar, 9/23/2025, 6:03 AM
Modified By	Swayam Tamrakar, 9/23/2025, 6:03 AM

Create Duplicate Rule

1. Setup → Quick Find → **Duplicate Rules** → **New Rule** → choose **Contact**.
2. Rule Name: Contact_Duplicate_Rule_Students.
3. Under **Matching Rule**, select the matching rule you created.
4. Action on create/edit:
 - **Alert** (allow save but show alert to user) or **Block** (prevent save) — choose based on policy.
5. Under **Actions**, specify what to do on create/edit — you can create a report, allow users to bypass with reason, or block.

6. Save and **Activate** the duplicate rule.

d

SETUP

Duplicate Rules

Contact Duplicate Rule

Contact_Duplicate_Rule_Students

[Back to List: Custom Apps](#)

[Help for this Page](#)

Duplicate Rule Detail

Edit

Delete

Clone

Deactivate

Rule Name	Contact_Duplicate_Rule_Students	Order	3 of 3 [Reorder]
Description			
Object	Contact		
Record-Level Security	Enforce sharing rules		
Action On Create	Allow	Operations On Create	☒ Alert ☒ Report
Action On Edit	Allow	Operations On Edit	☒ Alert ☒ Report
Alert Text	Use one of these records?		
Active	☒		
Matching Rule	☒ Contact Match By Email Or StudentID Mapped ☒	Matching Criteria	Contact: Email EXACT MatchBlank = FALSE
Conditions			
Created By	Swayam Tamrakar , 9/23/2025, 6:05 AM	Modified By	Swayam Tamrakar , 9/23/2025, 6:05 AM

Edit

Delete

Clone

Deactivate

Test

- Attempt to create a Contact with same email or Student_ID__c — verify the behavior (alert or block).

The screenshot shows a Salesforce contact creation form for a user named Swayam Tamrakar. The form includes fields for Name (Salutation, First Name, Last Name), Account Name (XYZ Engineering College), Title, Department, Birthdate, Reports To, Lead Source, Important, GPA, Home Phone, Mobile, Other Phone, Fax, Email (epic.orgfarm@salesforce.com), Assistant, and Asst. Phone. A red error message box is displayed in the center, stating: "We hit a snag. You can't save this record because a duplicate record already exists. To save, use different information. View Duplicates". The form is partially filled out, and the "Save" button is visible at the bottom right.

8.3 Backups — Data Export (manual & scheduled)

Manual Export (Developer Org)

1. Setup → Quick Find → **Data Export** → Click **Export Now**.
2. Choose objects to export (or select **Include all data**). For faster test, select Accounts, Contacts, Internship__c, Student_Skill__c, etc.

3. Click **Start Export**. Salesforce prepares a ZIP file and sends email when available. Download the ZIP and store it securely.

The screenshot shows the 'Data Export' setup page in Salesforce. The main heading is 'Monthly Export Service'. Below it, a message states: 'Data Export lets you prepare a copy of all your data in salesforce.com. From this page you can start the export process manually or schedule it to run automatically. When an export is ready for download you will receive an email containing a link that allows you to download the file(s). The export files are also available on this page for 48 hours, after which time they are deleted.'

A yellow box indicates 'Next scheduled export: None'. Below this are buttons for 'Export Now' and 'Schedule Export'.

Export details are listed:

- Scheduled By: Swagyan Tamrakar
- Schedule Date: 9/23/2025
- Export File Encoding: ISO-8859-1 (General US & Western European, ISO-LATIN-1)

A table shows the export file details:

Action	File Name	File Size
download	WE_00Dgl000007QLcfUAG_1.ZIP	254.5K

Scheduled Export (if available in your edition)

1. On **Data Export** page → **Schedule Export** (if enabled): choose **Weekly** or **Monthly**, select day/time and objects. Save.

The screenshot shows the 'Schedule Data Export' page. It includes a 'Save' and 'Cancel' button at the top. The 'Export File Encoding' is set to 'ISO-8859-1 (General US & Western European, ISO-LATIN-1)'. There are checkboxes for 'Include images, documents, and attachments', 'Include Salesforce Files and Salesforce Chatter Content document versions', and 'Replace carriage returns with spaces'. The 'Schedule Data Export' checkbox is checked.

The 'Frequency' section shows 'On day 1 of every month' selected. The 'Start' date is 9/24/2025 and the 'End' date is 10/24/2025. The 'Preferred Start Time' is set to 'None'.

The 'Exported Data' section allows selecting what type of information to include. The 'Include all data' checkbox is checked. Below it, a grid of checkboxes lists various data types: Contract, Asset, Lead, BusinessProcess, Campaign, CaseContactRole, Order, Account, Partner, NotificationMember, CampaignMember, CaseHistory2, OrderItem, Contact, Product2, UserRole, Case, and CaseSolution.

Tip: Always download and keep backup before any mass delete or destructive deploy.

Phase 9: Reporting, Dashboards & Security Review

9.1. Create a Report Folder

1. App Launcher → **Reports** → click **All Folders** (left pane).
2. Click **New Folder**.
 - o Name: Student Success Reports
3. Click **Save**.

Why: keeps reports & dashboards secure and easy to share.

The screenshot shows the 'Reports' section of the Student Success application. The 'All Folders' view is active, displaying a table of reports and folders. The table has columns for Name, Created By, Created On, Last Modified By, and Last Modified Date. The 'Student Success Reports' folder is highlighted in the left sidebar and the table.

REPORTS	Name	Created By	Created On	Last Modified By	Last Modified Date
Recent	Einstein Bot Reports	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Created by Me	Einstein Bot Reports Spring '23	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Private Reports	Einstein Bot Reports Summer '23	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Public Reports	Einstein Bot Reports Summer '22	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
All Reports	Einstein Bot Reports Winter '23	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
FOLDERS					
All Folders	Enablement Dashboard Reports Spring '24	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Created by Me	Enablement Dashboard Reports Summer '24	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Shared with Me	PRM Reports	Automated Process	9/11/2025, 12:06 PM	Automated Process	9/11/2025, 12:06 PM
FAVORITES					
All Favorites	Student Success Reports	Swayam Tamrakar	9/24/2025, 10:58 AM	Swayam Tamrakar	9/24/2025, 10:58 AM
	TCS Report Folder	Swayam Tamrakar	8/13/2025, 7:19 AM	Swayam Tamrakar	8/13/2025, 7:19 AM

9.2. Create Custom Report Types

Contacts with Internships

1. Setup (⚙️) → **Quick Find** → **Report Types** → **New Custom Report Type**.
2. Fill:
 - o **Primary Object:** Contacts
 - o **Report Type Label:** Students with Internships
 - o **Report Type Name:** Students_with_Internships
 - o **Category:** Contacts & Accounts (or Other Reports)

- **Deployment Status: Deployed**
 - Description: Students and their internship records
3. Click **Next** → On the next screen click **Click to relate another object** and pick your custom object Internship__c.
 4. Choose relationship: **Each "A" record must have at least one "B" record** (choose *with* or *without* if you want students even without internships).
 5. Save.

The screenshot shows the Salesforce Setup interface. The left sidebar contains a search bar with 'report' and a navigation menu with categories like Feature Settings, Analytics, Reports & Dashboards, Report Types, Reporting Snapshots, Reports and Dashboards Settings, Security, and Guest User Sharing Rule Access Report. The main content area is titled 'Students with Internships' and includes buttons for 'Preview Layout', 'Edit Layout', 'Clone', and 'Delete'. Below the title, there is a 'Close' button and a description: 'Below is the information for this custom report type. You can click the buttons on this to preview or update information for the custom report type'.

The 'Details' section shows the following information:

- Display Name:** Students with Internships
- API Name:** Students_with_Internships
- Description:** Students and their internship records
- Created By:** Swayam Tamrakar, 9/24/25, 11:33 PM
- Store in:** accounts
- Deployed:** Deployed
- Modified By:** Swayam Tamrakar, 9/24/25, 11:33 PM

The 'Object Relationships' section shows a Venn diagram with two overlapping circles, A and B. Circle A is labeled 'Contacts (A)' and circle B is labeled 'Internships'. The text below the diagram says '.....with at least one related record from Interns'. Below the diagram is a small table with two columns, A and B, and two rows of data.

The 'Fields' section shows a table with the following data:

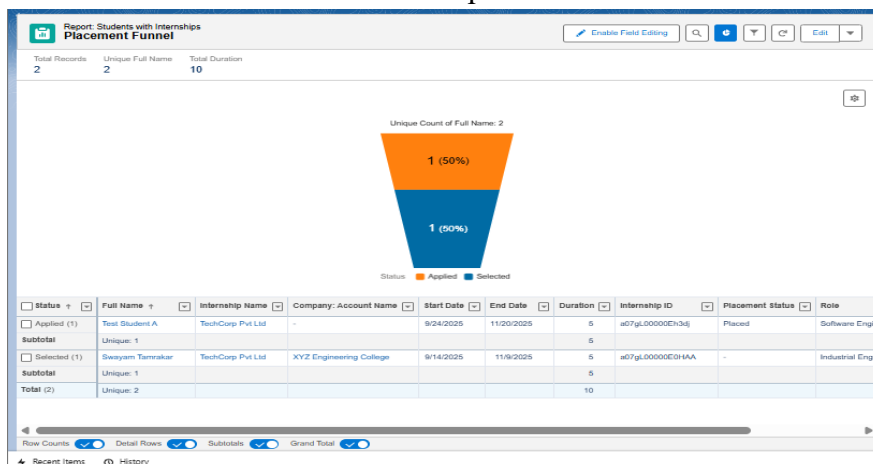
Source Object	Included Fields
Contacts	77
Internships	16

9.3. Create Reports

A — Placement Funnel

App Launcher → **Reports** → **New Report**.

1. In the report type chooser select: **Students with Internships** (the custom report type you created). Click **Continue**.
2. Filters:
 - **Show:** All students (or specify Record Type = Student)
 - **Date Range:** All Time (or internship Start Date filter)
 - Add filter: Internship__c.Status__c (or Offer_Status__c) **is not empty**.
3. Grouping:
 - Drag the Internship: Status (Offer_Status__c) to **Group Rows**. This will create rows per status (Applied, Interviewing, Selected, Completed, Rejected).
4. Columns: keep Student Name, Company (Account), Start Date.
5. Summaries:
 - Click the dropdown on Student Name column → **Summarize** → select **Count**. This gives count of students per status.
6. Add Chart (within report):
 - Click the chart icon (top right of the report).
 - Chart Type: **Funnel**
7. Save → Folder: Student Success Reports → Name: Placement Funnel.



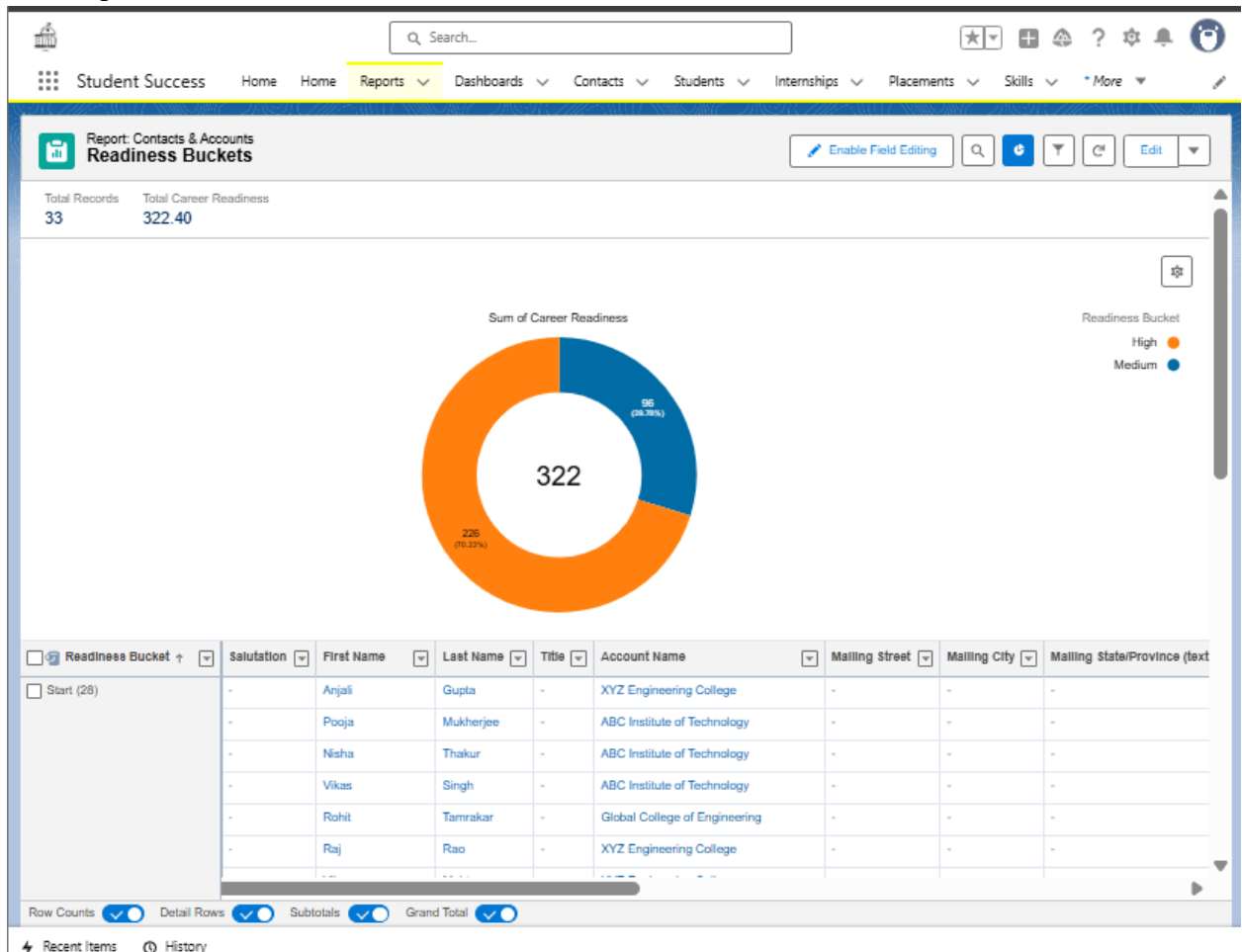
B—Average Career Readiness

1. New Report → choose **Contacts** (or custom report type that includes Career_Readiness__c).
2. Filters: Record Type = Student.
3. Add column: Career_Readiness__c (numeric).
4. Click column dropdown → **Summarize** → choose **Average**. (This computes avg readiness.)
5. Remove row grouping (leave as tabular or summary with no grouping).
6. Save → Name: Avg Career Readiness.

[illegible]

9.4. Use Bucket Fields to create Readiness Buckets

1. Edit Avg Career Readiness report or create a new Contacts report with Career_Readiness__c.
2. In the Columns header, click ▼ → **Add Bucket Column**.
 - Bucket Field Source: Career_Readiness__c.
 - Create buckets:
 - Low (0-30) → 0 to 30
 - Medium (31-60) → 31 to 60
 - High (61-100) → 61 to 100
3. Save the bucket as Readiness Bucket.
4. Add a chart: Chart Type **Donut**, Grouping: Readiness Bucket, Metric: Record Count.
5. Save report: Readiness Buckets.

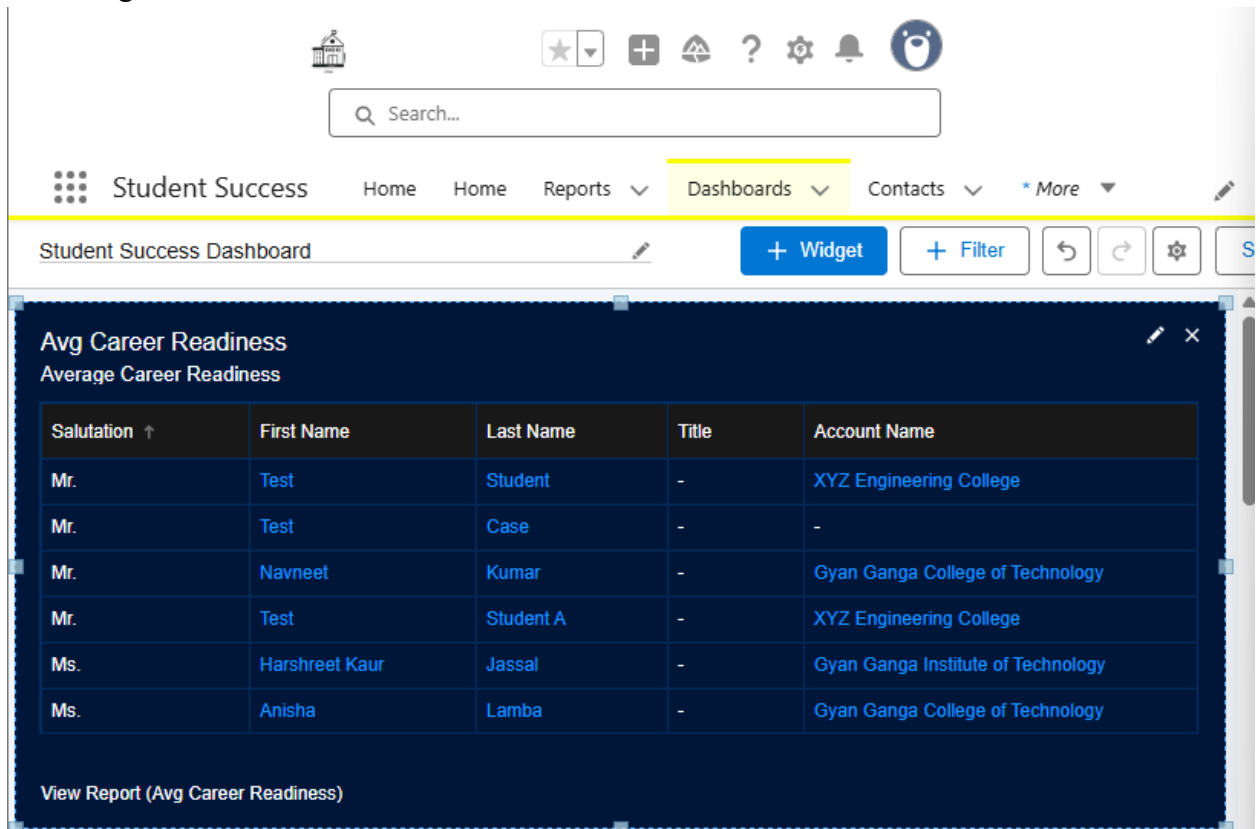


9.5. Build Dashboard & Add Components

1. App Launcher → **Dashboards** → **New Dashboard**.
 - Dashboard Name: Student Success Dashboard
 - Folder: Student Success Reports
 - Click **Create**.

A — Add Table: Average Career Readiness

1. Click + **Component**.
2. Select Report: Avg Career Readiness (the report with Average summary).
3. Select **Table** component type.
4. Data: choose Metric = Average of Career_Readiness__c.
5. Title: Avg Career Readiness. Click **Add**.



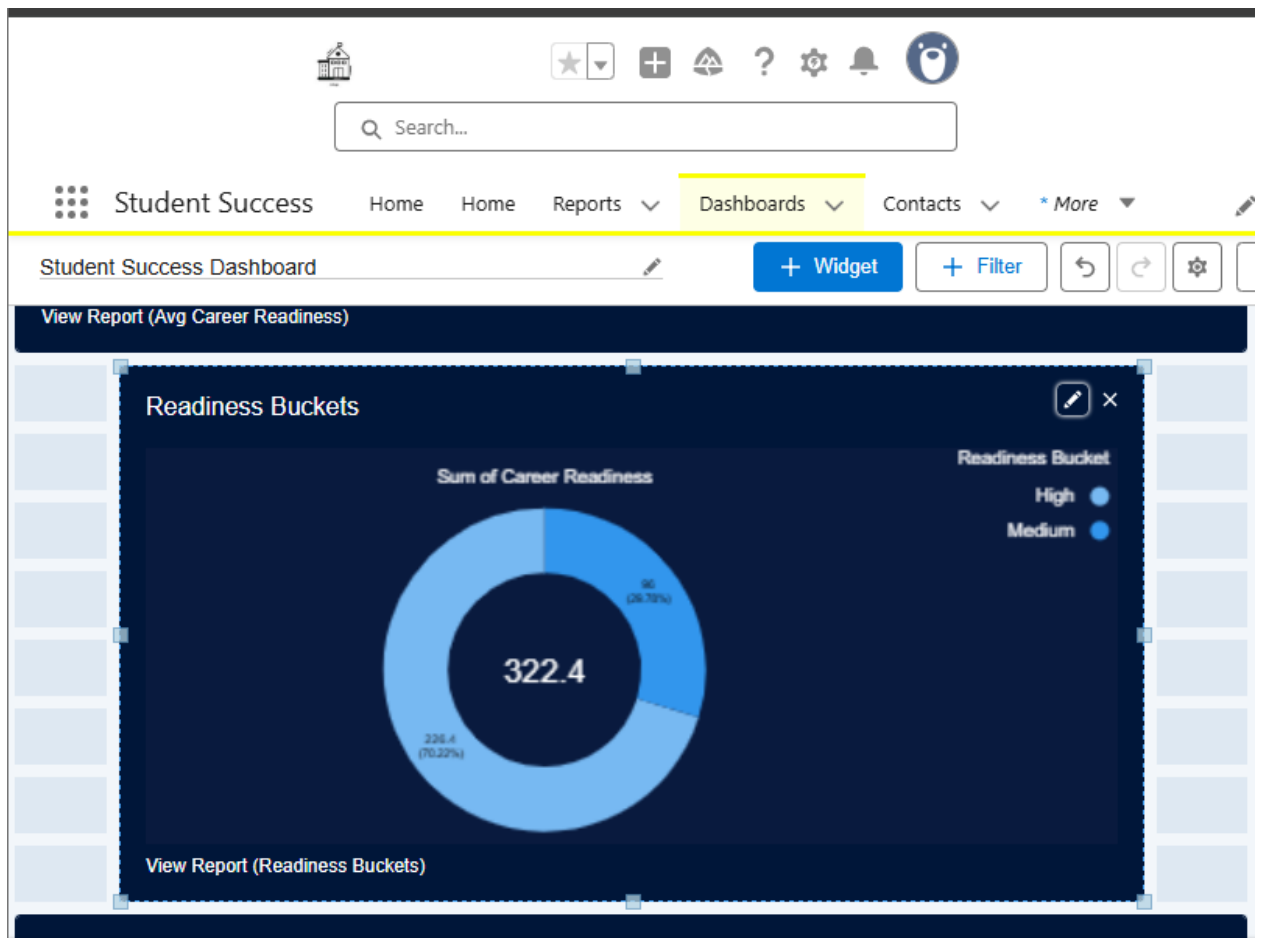
The screenshot displays the Salesforce interface for the 'Student Success Dashboard'. The dashboard title is 'Student Success Dashboard'. The main component is a table titled 'Avg Career Readiness' with the subtitle 'Average Career Readiness'. The table has five columns: Salutation, First Name, Last Name, Title, and Account Name. The data rows are as follows:

Salutation	First Name	Last Name	Title	Account Name
Mr.	Test	Student	-	XYZ Engineering College
Mr.	Test	Case	-	-
Mr.	Navneet	Kumar	-	Gyan Ganga College of Technology
Mr.	Test	Student A	-	XYZ Engineering College
Ms.	Harshreet Kaur	Jassal	-	Gyan Ganga Institute of Technology
Ms.	Anisha	Lamba	-	Gyan Ganga College of Technology

Below the table, there is a link labeled 'View Report (Avg Career Readiness)'.

B — Add Donut: Readiness Buckets

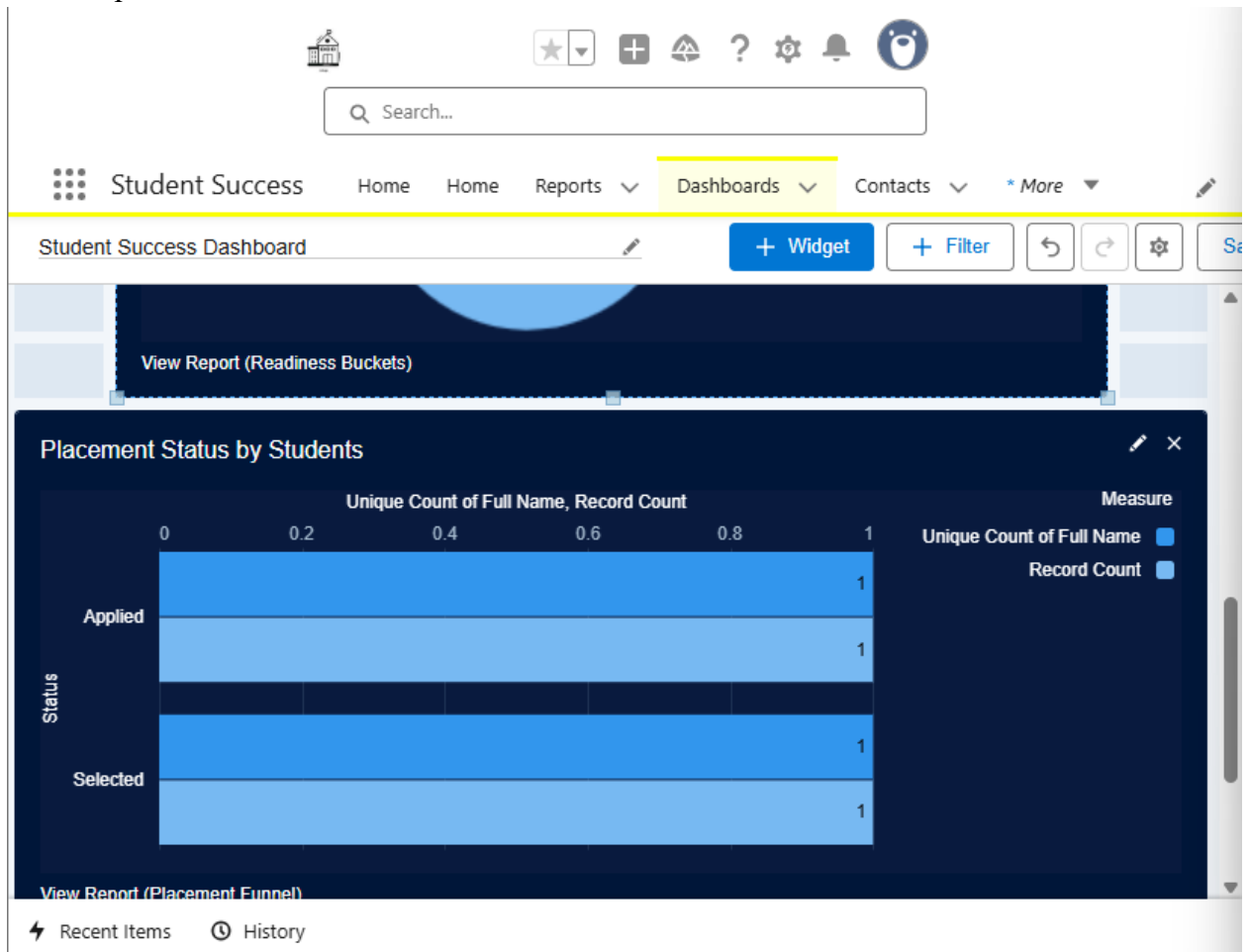
1. + **Component** → select report Readiness Buckets.
2. Component type: **Donut**.
3. Configure to show Readiness Bucket groups and set labels/legend.
4. Title: Readiness Distribution. Click **Add**.



C — Add Bar: Placement

1. + **Component** → select Top Skills by Department report.
2. Component type: **Bar Chart** (or horizontal bar).
3. In component properties— choose Top 10 by Record Count.

4. Title: Top Skills → Add.



D — Arrange & Save

1. Drag components to arrange.
2. Click **Save** → **Done**.

Tip: in Dashboard component settings you can choose to display most recent data, enable "Show as table" or "Show labels".

9.6. Make Dashboard Dynamic

1. Open the Dashboard → click **Edit** → Click the gear (component run settings) or top-right **View Dashboard As**.
2. Choose **The dashboard viewer**Save.

The screenshot shows a 'Properties' dialog box for a dashboard. It has a title bar 'Properties'. Inside, there's a 'Name' field with the text 'Student Success Dashboard'. Below it is a 'Description' field. Then a 'Folder' field with the text 'Student Success Reports' and a 'Select Folder' button. Below that, it says 'This dashboard is owned by Swayam Tamrakar'. Under the heading 'View Dashboard As', there are four radio button options: 'Me', 'Another person', 'The dashboard viewer' (which is selected), and 'Let dashboard viewers choose whom they view the dashboard as'. At the bottom right, there are 'Cancel' and 'Save' buttons.

Note: dynamic dashboards are useful so Placement Officers see only their dept's students.

9.7. Sharing & Folder Access

1. In Reports app → locate Student Success Reports folder → Down arrow → **Share**.

The screenshot shows the 'Reports' app interface with the 'All Folders' view. A table lists various folders and reports. A context menu is open for the 'Student Success Reports' folder, showing options: Favorite, Share, Rename, and Delete.

	Name	Created By	Created On	Last Modified By	Last Modified Date
REPORTS					
Recent	Einstein Bot Reports	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Created by Me	Einstein Bot Reports Spring '23	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Private Reports	Einstein Bot Reports Summer '23	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Public Reports	Einstein Bot Reports Summer '22	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
All Reports	Einstein Bot Reports Winter '23	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
FOLDERS					
All Folders	Enablement Dashboard Reports Spring '24	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Created by Me	Enablement Dashboard Reports Summer '24	Automated Process	7/17/2025, 12:07 AM	Automated Process	7/17/2025, 12:07 AM
Shared with Me					
FAVORITES					
All Favorites	PRM Reports	Automated Process	9/11/2025, 12:06 PM	Automated Process	9/11/2025, 12:06 PM
	Student Success Reports	Swayam Tamrakar	9/24/2025, 10:58 AM	Swayam Tamrakar	9/24/2025, 10:58 AM
	TCS Report Folder	Swayam Tamrakar	8/13/2025, 7:19 AM	Swayam Tamrakar	8/13/2025, 7:19 AM

2. Add Roles, Users, or Public Groups (e.g., Placement Officers) and set access level (Viewer/Editor). Save.

The 'Share folder' dialog box shows the following settings:

- Share With:** Public Groups
- Names:** Placement Officers
- Access:** Manage
- Who Can Access:** Swayam Tamrakar (Users)

The 'Done' button is visible at the bottom right.

9.8. Security Review

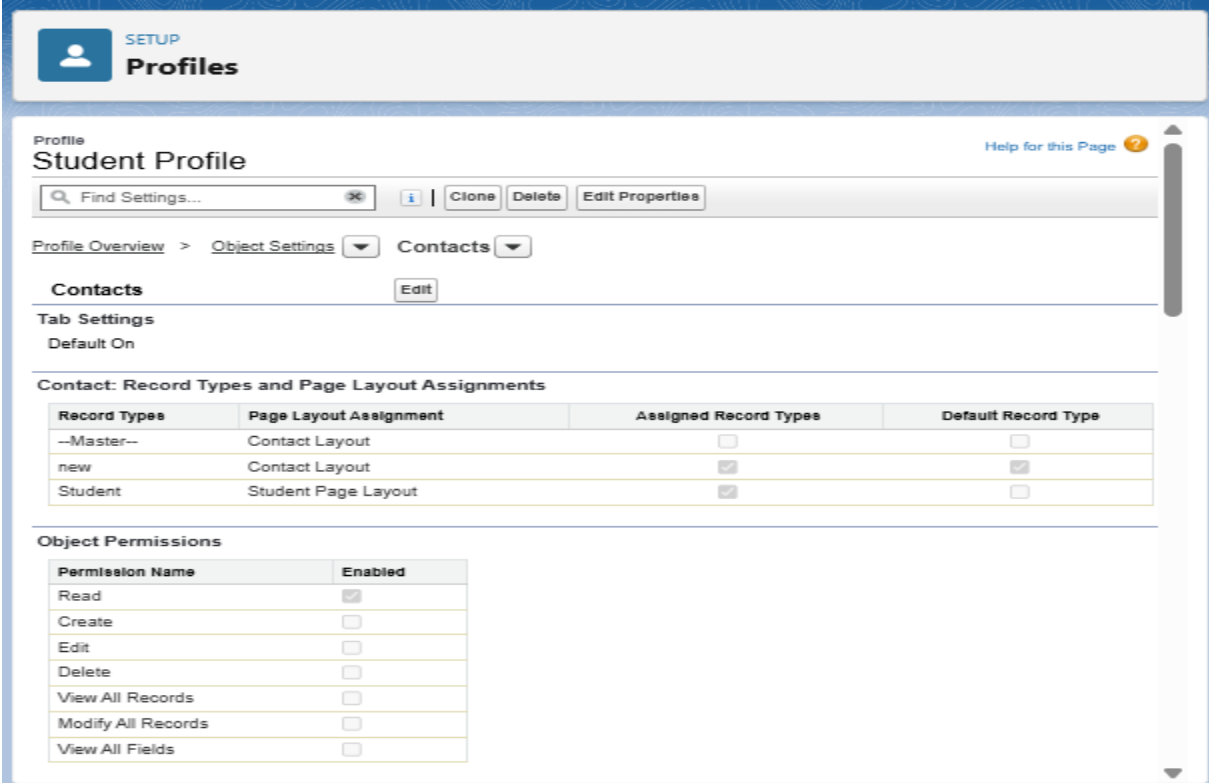
9.81 Field-Level Security (FLS) — hide SSN__c or other PII

1. Setup → **Object Manager** → **Contact** (or object with SSN field).
2. Fields & Relationships → find SSN__c → click field label → **Set Field-Level Security**.
3. Uncheck **Visible** for the Student_Profile (and any other profile that should not see SSN).
4. Save.
5. Also remove SSN__c from the Student Page Layout: Object Manager → Contact → Page Layouts → Edit Student Page Layout → remove SSN__c field → Save.

Test: Login as a Student user (or use “Login” as that user) and verify SSN is not visible.

9.82 Profiles & Permission Sets — least privilege

1. Setup → **Profiles** → open Student_Profile.
 - Click **Object Settings** → **Contacts** → Edit → ensure students have **Read** on their own contact only (use Sharing to limit edit). Remove unnecessary object permissions.



The screenshot shows the Salesforce Setup interface for the 'Student Profile'. The 'Object Settings' tab is selected, and the 'Contacts' object is chosen. The 'Edit' button is visible. Below the 'Tab Settings' (Default On), the 'Contact: Record Types and Page Layout Assignments' table is displayed. The table has four columns: Record Types, Page Layout Assignment, Assigned Record Types, and Default Record Type. The 'Student' record type is assigned the 'Student Page Layout'. Below this, the 'Object Permissions' table is shown, listing various permissions with checkboxes for enabling them. The 'Read' permission is currently checked.

Record Types	Page Layout Assignment	Assigned Record Types	Default Record Type
--Master--	Contact Layout	☐	☐
new	Contact Layout	☒	☒
Student	Student Page Layout	☒	☐

Permission Name	Enabled
Read	☒
Create	☐
Edit	☐
Delete	☐
View All Records	☐
Modify All Records	☐
View All Fields	☐

2. Setup → **Permission Sets** → create Placement_Officer_PS for extra rights (e.g., Export Reports).

The screenshot shows the 'Permission Sets' configuration page for 'Placement_Officer_PS'. The page has a header with a 'SETUP' icon and the title 'Permission Sets'. Below the header, there's a search bar 'Find Settings...' and buttons for 'Clone', 'Delete', 'Edit Properties', 'Manage Assignments', and 'View Summary'. The main content area is divided into sections: 'Permission Set Overview' (with a dropdown for 'Object Settings' and 'Placements'), 'Placements' (with an 'Edit' button), 'Tab Settings' (with 'Available' and 'Visible' checkboxes), 'Object Permissions' (a table with columns 'Permission Name' and 'Enabled'), and 'Field Permissions' (a table with columns 'Field Name', 'Field API Name', 'Read Access', and 'Edit Access').

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☒
View All Records	☐
Modify All Records	☐
View All Fields	☐

Field Name	Field API Name	Read Access	Edit Access
Company	Company__c	☒	☒
Company Name	Company_Name__c	☒	☒

- Assign the permission set to users: Permission Set → Manage Assignments → Add Assignments.

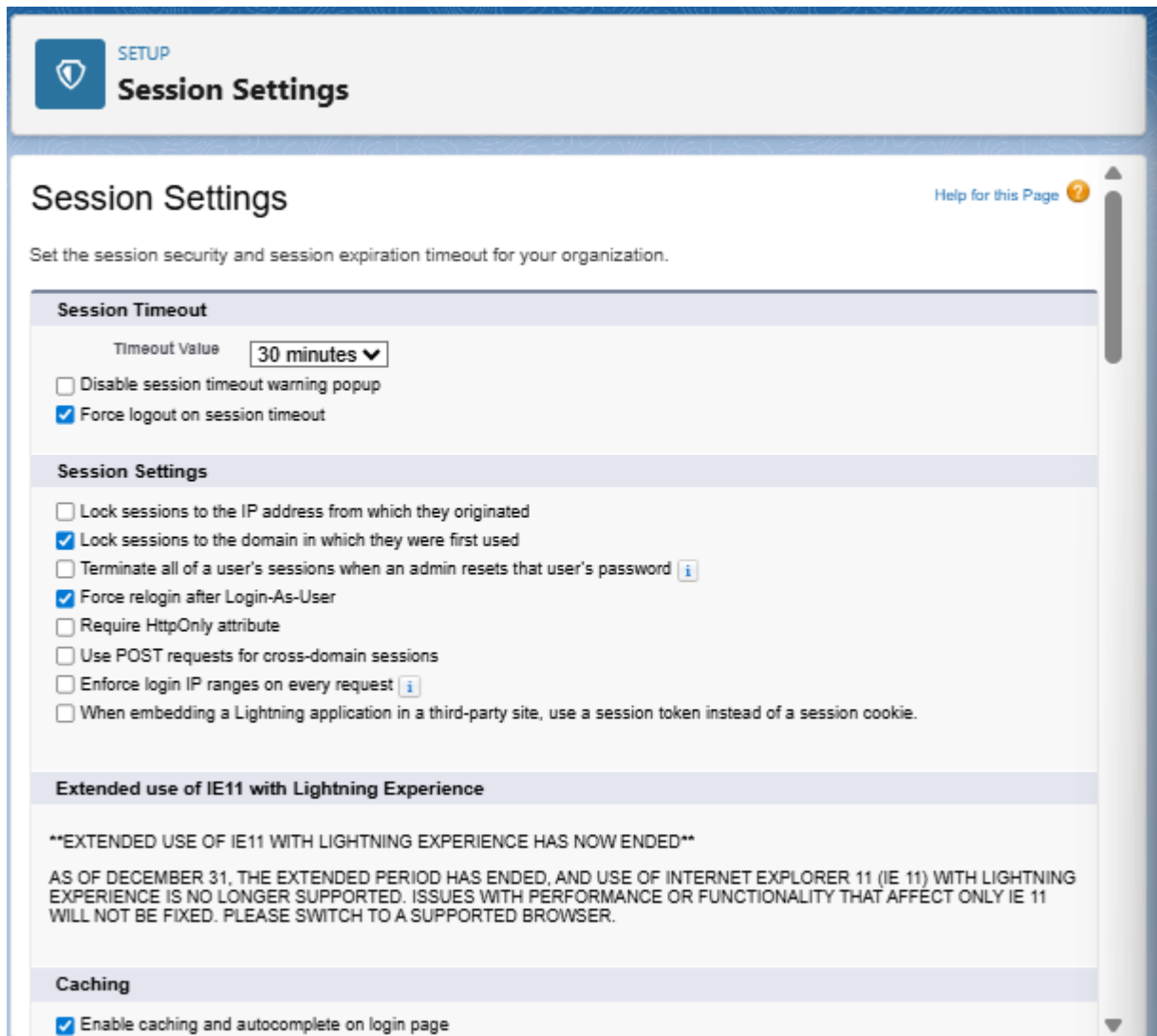
The screenshot shows the 'Permission Sets' configuration page for 'Placement Officer'. The page has a header with a 'SETUP' icon and the title 'Permission Sets'. Below the header, there's a search bar 'Find Settings...' and buttons for 'Clone', 'Edit Properties', 'Manage Assignments', and 'View Summary'. The main content area is divided into sections: 'Permission Set Overview' (with a dropdown for 'System Permissions'), 'System Permissions' (with an 'Edit' button), and 'System' (a table with columns 'Permission Name', 'Enabled', and 'Description').

Permission Name	Enabled	Description
Access Actionable Relationship Center for partner users	☐	Grants access to ARC Relationship Graph, ARC Highlights Panel, and ARC Record Details Lightning components to partner users.
Access Activities	☐	Access tasks, events, calendar, and email.
Access Agent for Setup	☐	Use agents to ask for help in admin-related tasks.
Access Banking Relationship Assistance	☐	Enable Banking Relationship Assistance.
Access Banking Service Agent	☐	Grants users access to Banking Service Assistance AI agent.
Access Customer Asset Lifecycle Management APIs	☐	Use APIs to manage lifecycle-managed assets.
Access Custom Mobile Apps	☐	Allow user to run custom mobile apps.
Access Data Cloud Data Explorer	☐	Allows user access Data Cloud Data Explorer.
Access Data Cloud Profile Explorer	☐	Allows user access Data Cloud Profile Explorer.
Access drag-and-drop content builder	☐	Create and manage email templates and content in a drag-and-drop builder.
Access Experience Management	☐	Access pages and dashboards available in Experience Management.
Access Libraries	☐	Access libraries.

Test: Log in as Placement Officer and Student to validate each sees appropriate data.

9.83 Session Settings & Login IP Ranges

1. Setup → **Session Settings** → set **Session timeout** to reasonable value (e.g., 30 minutes). Save.



The screenshot shows the 'Session Settings' page in a web application. At the top, there is a 'SETUP' button and the title 'Session Settings'. Below the title, there is a subtitle 'Session Settings' and a 'Help for this Page' link. The main content area is divided into several sections:

- Session Timeout**: This section contains a 'Timeout Value' dropdown menu set to '30 minutes'. Below it, there are two checkboxes: 'Disable session timeout warning popup' (unchecked) and 'Force logout on session timeout' (checked).
- Session Settings**: This section contains several checkboxes: 'Lock sessions to the IP address from which they originated' (unchecked), 'Lock sessions to the domain in which they were first used' (checked), 'Terminate all of a user's sessions when an admin resets that user's password' (unchecked), 'Force relogin after Login-As-User' (checked), 'Require HttpOnly attribute' (unchecked), 'Use POST requests for cross-domain sessions' (unchecked), 'Enforce login IP ranges on every request' (unchecked), and 'When embedding a Lightning application in a third-party site, use a session token instead of a session cookie' (unchecked).
- Extended use of IE11 with Lightning Experience**: This section contains a message: '**EXTENDED USE OF IE11 WITH LIGHTNING EXPERIENCE HAS NOW ENDED**'. Below this, it states: 'AS OF DECEMBER 31, THE EXTENDED PERIOD HAS ENDED, AND USE OF INTERNET EXPLORER 11 (IE 11) WITH LIGHTNING EXPERIENCE IS NO LONGER SUPPORTED. ISSUES WITH PERFORMANCE OR FUNCTIONALITY THAT AFFECT ONLY IE 11 WILL NOT BE FIXED. PLEASE SWITCH TO A SUPPORTED BROWSER.'
- Caching**: This section contains one checkbox: 'Enable caching and autocomplete on login page' (checked).

2. Setup → **Network Access** → **New** → add **Trusted IP Ranges** for your org (optional) — this prevents logins from outside ranges without verification.

The screenshot shows the Salesforce Setup interface. The top navigation bar includes the Salesforce logo, a search bar labeled "Search Setup", and several utility icons. Below this, the "Setup" menu is open, showing "Home" and "Object Manager". The left sidebar contains a search bar with "netwo" and a "Security" section with "Network Access" highlighted. The main content area is titled "Network Access" and includes a "Help for this Page" link. A descriptive paragraph explains that the list contains IP address ranges from trusted sources. Below this is a table titled "Trusted IP Ranges" with a "New" button. The table has four columns: "Action", "Start IP Address", "End IP Address", and "Description". One entry is shown with the start IP "192.168.1.1", end IP "192.168.1.255", and description "Office Network Range".

Search Setup

Setup Home Object Manager

netwo

Security

Network Access

Didn't find what you're looking for? Try using Global Search.

Network Access

Help for this Page

The list below contains IP address ranges from sources that your organization trusts. Users logging in to salesforce.com with a browser from trusted networks are allowed to access salesforce.com without having to activate their computers.

Action	Start IP Address	End IP Address	Description
Edit Del	192.168.1.1	192.168.1.255	Office Network Range

3. For admin profiles: Setup → **Profiles** → select System Administrator → **Login IP Ranges** → Add allowed IPs to restrict admin console access.

The screenshot displays the Cisco ISE Setup web interface. The top navigation bar includes a search box labeled 'Search Setup' and various utility icons. The left sidebar shows the 'Setup' menu with 'Home' and 'Object Manager' options. The 'Profiles' section is highlighted in the sidebar, and a search for 'profile' is visible. The main content area shows the 'Sales' profile configuration. The 'Login IP Ranges' tab is selected, displaying a table with one entry: 'Admin Allowed Range' with IP start address 192.168.1.1 and IP end address 192.168.1.255. The table has columns for 'Action', 'IP Start Address', 'IP End Address', and 'Description'. The 'Action' column contains links for 'Edit' and 'Delete'. An 'Add IP Ranges' button is located above the table.

Profile: Sales

Find Settings... Clone Delete Edit Properties

Profile Overview > Login IP Ranges

Login IP Ranges Add IP Ranges

Action	IP Start Address	IP End Address	Description
Edit Delete	192.168.1.1	192.168.1.255	Admin Allowed Range

9.84 Password Policies

1. Setup → **Password Policies** (under Security Controls) → set **Password Expiration**, **Complexity** rules.

The screenshot shows the Microsoft 365 Setup interface. The top navigation bar includes the Microsoft logo, a search bar labeled 'Search Setup', and several utility icons. Below this, the 'Setup' tab is active, with sub-tabs for 'Home' and 'Object Manager'. A left-hand navigation pane contains a search bar with 'pass' entered, a 'Security' section with 'Expire All Passwords' and 'Password Policies' (highlighted), and a message: 'Didn't find what you're looking for? Try using Global Search.' The main content area is titled 'Password Policies' with a subtitle 'Set the password restrictions and login lockout policies for all users.' It features a list of settings, each with a red exclamation mark icon indicating required information. The settings are: 'User passwords expire in' (90 days), 'Enforce password history' (3 passwords remembered), 'Minimum password length' (8), 'Password complexity requirement' (Must include numbers, uppercase and lowercase letters, and special characters), 'Password question requirement' (Cannot contain password), 'Maximum invalid login attempts' (10), 'Lockout effective period' (15 minutes), 'Obscure secret answer for password resets' (unchecked), 'Require a minimum 1 day password lifetime' (unchecked), and 'Allow use of setPassword() API for self-resets' (checked). A vertical scrollbar is on the right side of the settings list.

Setup

Search Setup

Setup Home Object Manager

pass

Security

Expire All Passwords

Password Policies

Didn't find what you're looking for?
Try using Global Search.

SETUP

Password Policies

Set the password restrictions and login lockout policies for all users.

Help for this Page

Password Policies

! = Required Information

- User passwords expire in: 90 days
- Enforce password history: 3 passwords remembered
- Minimum password length: 8
- Password complexity requirement: Must include numbers, uppercase and lowercase letters, and special characters
- Password question requirement: Cannot contain password
- Maximum invalid login attempts: 10
- Lockout effective period: 15 minutes
- Obscure secret answer for password resets: ☐
- Require a minimum 1 day password lifetime: ☐
- Allow use of setPassword() API for self-resets: ☒

Forgot Password / Locked Account Assistance

9.85 Audit Trail & Login History

1. Setup → **View Setup Audit Trail** → review recent configuration changes.
 - Click **Download** to export last 180 days.

The screenshot shows the Salesforce Setup interface. A browser window displays the URL `orgfarm-081ad31f5a-dev-ed.develop.lightning.force.com/lightni...`. A download notification for `SetupAuditTrail1758799180560.csv` (111 KB) is visible. The Setup menu is open, and the **View Setup Audit Trail** option under the Security section is selected.

The **View Setup Audit Trail** page displays a table of configuration changes:

Time	Username	Change	Object	User
9/25/2025, 4:10:55 AM PDT	cse22_swayamtamrakar887@agentforce.com	Permission set Placement Officer: assigned to user Prashant Kotha (UserID: [005gL000005AOxR])	Manage Users	
9/25/2025, 4:10:55 AM PDT	cse22_swayamtamrakar887@agentforce.com	Permission set Placement Officer: assigned to user OrgFarm EPIC (UserID: [005gL000004rFoD])	Manage Users	
9/25/2025, 4:08:20 AM PDT	placement1@demo.com	Logged out using Login-As access for Sethi	Manage Users	cse22_swayam
9/25/2025, 4:04:53 AM PDT	placement1@demo.com	Logged in using Login-As access for Sethi	Manage Users	cse22_swayam
9/25/2025, 4:01:28 AM PDT	cse22_swayamtamrakar887@agentforce.com	Changed profile Custom: Sales Profile: Student Success is now the default application	Manage Users	
9/25/2025, 4:00:34 AM PDT	cse22_swayamtamrakar887@agentforce.com	Changed profile Faculty Profile: Student Success is now the default application	Manage Users	
9/25/2025, 3:59:48 AM PDT	cse22_swayamtamrakar887@agentforce.com	Changed profile Placement Profile: Student Success is now the default application	Manage Users	

Download setup audit trail for last six months (Excel .csv file)

2. Setup → **Login History** → view user logins, failed logins, and IP addresses.

The screenshot displays the AWS IAM console's 'Login History' page. The left-hand navigation pane shows the 'Setup' menu with 'Home' and 'Object Manager' options. Below this, the 'Identity' section is expanded, and 'Login History' is selected. A search bar at the top of the console contains the text 'login h'. The main content area is titled 'Login History' and includes a 'Help for this Page' link. It provides instructions on downloading the last six months of login history, with a note that the location shown is approximate. Below the instructions are 'Download Options' for 'File Type' (CSV File selected, GZIP File unselected) and 'File Contents' (All Logins selected), with a 'Download Now' button. A table below shows a list of login events with columns for Username, Login Time, Source IP, Location, Login Type, Status, Browser, and Platform. The table lists six successful logins for the user 'cse22_swayamtamrakar887@agentforce.com' from India, all using Chrome 140 on Windows 10.

Search Setup

Setup Home Object Manager

login h

Identity

Login History

Didn't find what you're looking for?
Try using Global Search.

Login History

Download the last six months of login history. Or, filter and show up to 20,000 records of login history for the last six months.

Location shows the approximate location of the IP address from where the user logged in. To show more geographic information, such as approximate city and postal code, create a custom view to include those fields. Due to the nature of geolocation technology, the accuracy of geolocation fields (for example, country, city, postal code) may vary.

Download Options

File Type

☒ CSV File
☐ GZIP File

File Contents

All Logins

[Download Now](#)

View: All Create New View

Username	Login Time	Source IP	Location	Login Type	Status	Browser	Platform
cse22_swayamtamrakar887@agentforce.com	9/25/2025, 4:08:25 AM PDT	49.36.25.123	India	Application	Success	Chrome 140	Windows 10
cse22_swayamtamrakar887@agentforce.com	9/25/2025, 3:53:01 AM PDT	49.36.25.123	India	Application	Success	Chrome 140	Windows 10
cse22_swayamtamrakar887@agentforce.com	9/25/2025, 3:51:45 AM PDT	49.36.25.123	India	Application	Success	Chrome 140	Windows 10
cse22_swayamtamrakar887@agentforce.com	9/25/2025, 3:49:22 AM PDT	49.36.25.123	India	Application	Success	Chrome 140	Windows 10
cse22_swayamtamrakar887@agentforce.com	9/25/2025, 1:00:18 AM PDT	49.36.25.123	India	Application	Success	Chrome 140	Windows 10
cse22_swayamtamrakar887@agentforce.com	9/25/2025, 12:45:44 AM PDT	49.36.25.123	India	Application	Success	Chrome 140	Windows 10

3. Setup → **Apex Jobs** → check for background jobs running.

The screenshot shows the Salesforce Setup interface. On the left is a navigation menu with categories: Email, Custom Code, and Environments. Under Custom Code, 'Apex Jobs' is highlighted. Below the menu, a message says 'Didn't find what you're looking for? Try using Global Search.' The main content area is titled 'Apex Jobs' and includes a link to 'Click here to go to the new batch jobs page'. Below this is a green status bar indicating 'Percent of Asynchronous Apex Used: 0%' and a table of job execution details.

Apex Jobs

Monitor the status of all Apex jobs, and optionally, abort jobs that are in progress.

Percent of Asynchronous Apex Used: 0%
You have currently used 12 asynchronous Apex operations out of an allowed 24-hour organization limit of 250,000. To learn about Platform Apex Limits topic.

View: **All** [Create New View](#)

Action	Submitted Date	Job Type	Status	Status Detail	Total Batches	Batches Processed	Failures	Submitted By
	9/25/2025, 2:00 AM	Batch Apex	Completed	First error: Insert failed. First exception on row 0; first error: REQUIRED_FIELD_MISSING, Required fields are missing: [Deadline__c, Job_Post__c]: [Deadline__c, Job_Post__c]	3	3	3	Tamra Swaye
	9/25/2025, 2:00 AM	Batch Apex	Completed	First error: Insert failed. First exception on row 0; first error: REQUIRED_FIELD_MISSING, Required fields are missing: [Deadline__c, Job_Post__c]: [Deadline__c, Job_Post__c]	3	3	3	Tamra Swaye
	9/24/2025, 11:00 PM	Batch Apex	Completed	First error: Insert failed. First exception on row 0; first error: REQUIRED_FIELD_MISSING, Required fields are missing: [Deadline__c, Job_Post__c]: [Deadline__c, Job_Post__c]	3	3	3	Tamra Swaye
	9/24/2025, 2:00 AM	Batch Apex	Completed	First error: Insert failed. First exception on row 0; first error: REQUIRED_FIELD_MISSING, Required fields are missing: [Deadline__c, Job_Post__c]: [Deadline__c, Job_Post__c]	2	2	2	Tamra Swaye

9.86 Health Check

1. Setup → **Health Check** → run and review your org's security score.
2. Address *High Risk* and *Medium Risk* settings (click an item → *Fix* suggestions).

Search Setup

Setup Home Object Manager

health

Security

Health Check

Portal Health Check

Didn't find what you're looking for?
Try using Global Search.

SETUP Health Check

How well does your org meet Salesforce security standards? Reduce your security risk and limit data loss by optimizing the areas below.

Salesforce Baseline Standard ⌵ ⚙️ Fix Risks

95% Excellent 1

of the standard met

For more detail on logs and events, as well as retaining the history of your Health Check score, please reach out to your Salesforce Account team regarding Event Monitoring & Security Center.

High-Risk Security Settings (13)

STATUS	SETTING	GROUP	YOUR VALUE	STANDARD VALUE	ACTIONS
Warning	Number of Objects with Default External Access Set to Public	Sharing Settings	1	0	Edit ⚙️
Compliant	Expired Certificate	Certificate and Key Management	0	0	Edit ⚙️
	Number of security risk file	File Upload And	0 security risk file	0 security risk file	...

Phase 10: Quality Assurance Testing

- In Each Phases I included test case for every Salesforce feature you implement (e.g., Record Creations in objects, Approval Process, Automatic Task Creation, Flows, Triggers, Validation Rules, etc.).
 - Each test case includes:
 - ✓ Input details (what data/action you are testing).
 - ✓ Expected Output (what should happen).
 - ✓ Actual Output (with results).
 - Screenshots are also available for both input and output.
-

Demo-Video

Link:- https://drive.google.com/file/d/10XkP4gLe-Mx-WTmDHyneO_UUNI5biH-a/view?usp=sharing

Future Enhancements

To further improve the *Student Success & Career Guidance Portal*, the following enhancements can be implemented in future phases:

- **Chatbot Integration:**
Integrate an AI-powered chatbot (via Salesforce Einstein Bots or external API) to answer student queries instantly, such as internship deadlines, skill recommendations, or company details.
- **Advanced AI Recommendations:**
Extend the Career Recommendation engine with **Einstein AI / GPT-based suggestions** to provide personalized job and career guidance based on real-time student performance, skills, and job market trends.
- **Student Community Portal:**
Build a **Salesforce Experience Cloud** portal where students can log in, update skills, view recommended jobs, and track placement status directly.
- **Mobile App Integration:**
Enable access through Salesforce Mobile App for on-the-go tracking of student readiness and job applications.

- **Alumni Tracking Module:**
Add alumni profiles and integrate feedback loops, so institutions can track long-term career success and maintain strong industry connections.
 - **Third-Party LMS Integration:**
Integrate with popular Learning Management Systems (Moodle, Canvas, Blackboard) to fetch real-time course completion and skill achievements.
-

Conclusion

The *Student Success & Career Guidance Portal* built on Salesforce Education Cloud successfully addresses the challenges faced by colleges in tracking student performance, internships, and placements. By providing a **360° student view**, automating processes through **Flows** (such as the Job Sync Flow), and enabling **analytics via dashboards and reports**, the solution streamlines placement operations and supports data-driven decision-making.

The project demonstrates how Salesforce can be customized for the education sector by leveraging **standard objects, custom objects, automation, and security features**. It also ensures scalability, as future enhancements like **chatbot integration, AI-driven recommendations, and community portals** can be added without disrupting the existing system.

In conclusion, this CRM project not only improves institutional efficiency but also enhances student employability by aligning academic progress with career opportunities. It stands as a strong example of how Salesforce can transform higher education management and placement processes.
